

Unsupervised Deep Multi-Shape Matching

ECCV 2022

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Presented by : **Veerav Chebrolu**

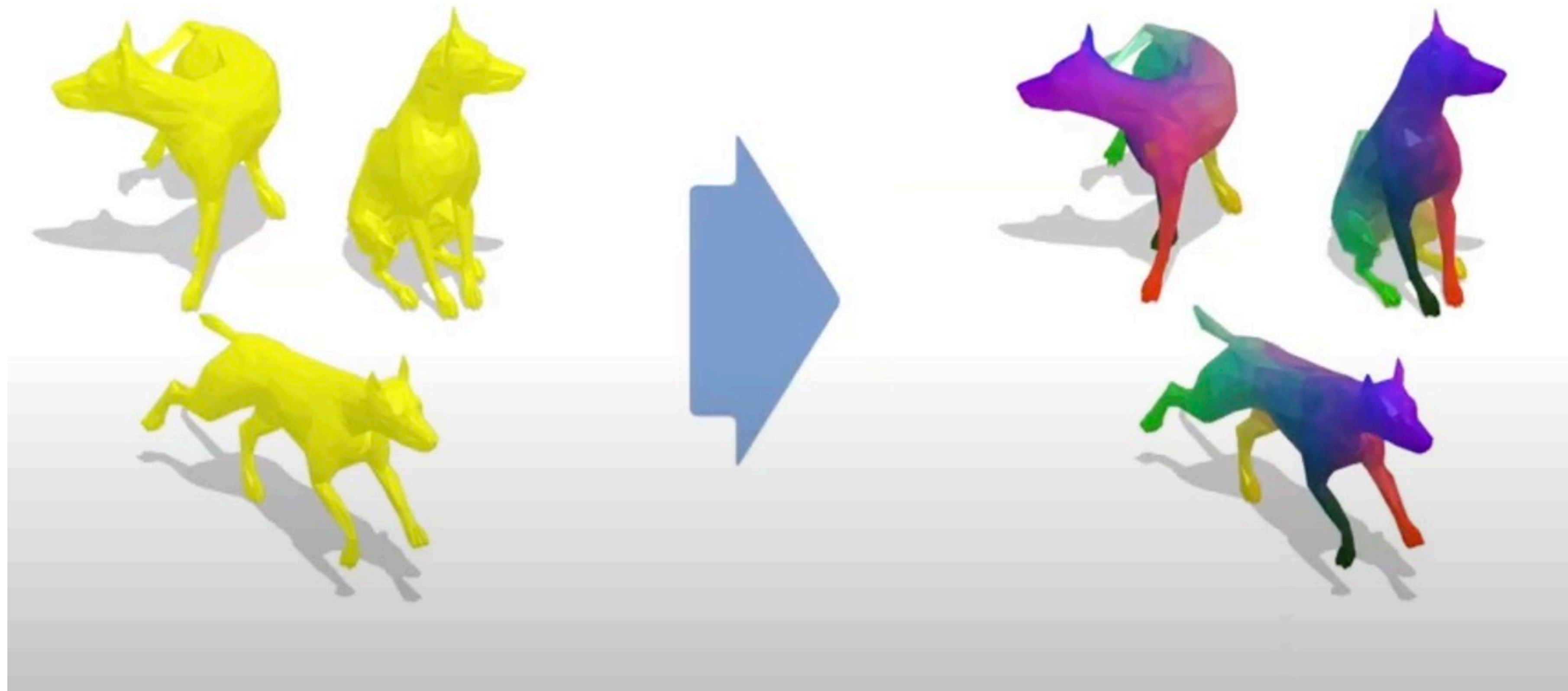
Date : **12 July 2023**

M.Sc. Scientific Computing

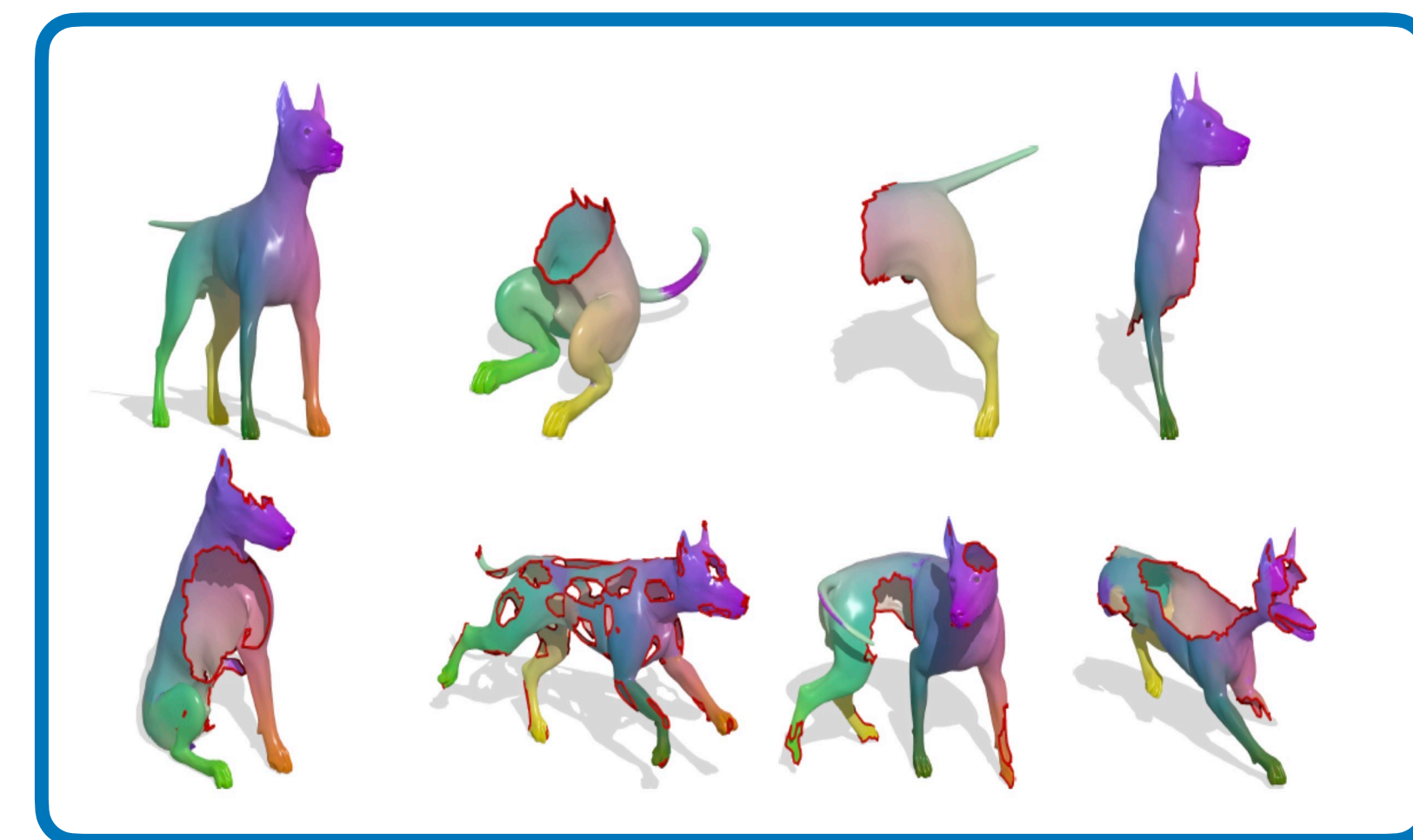
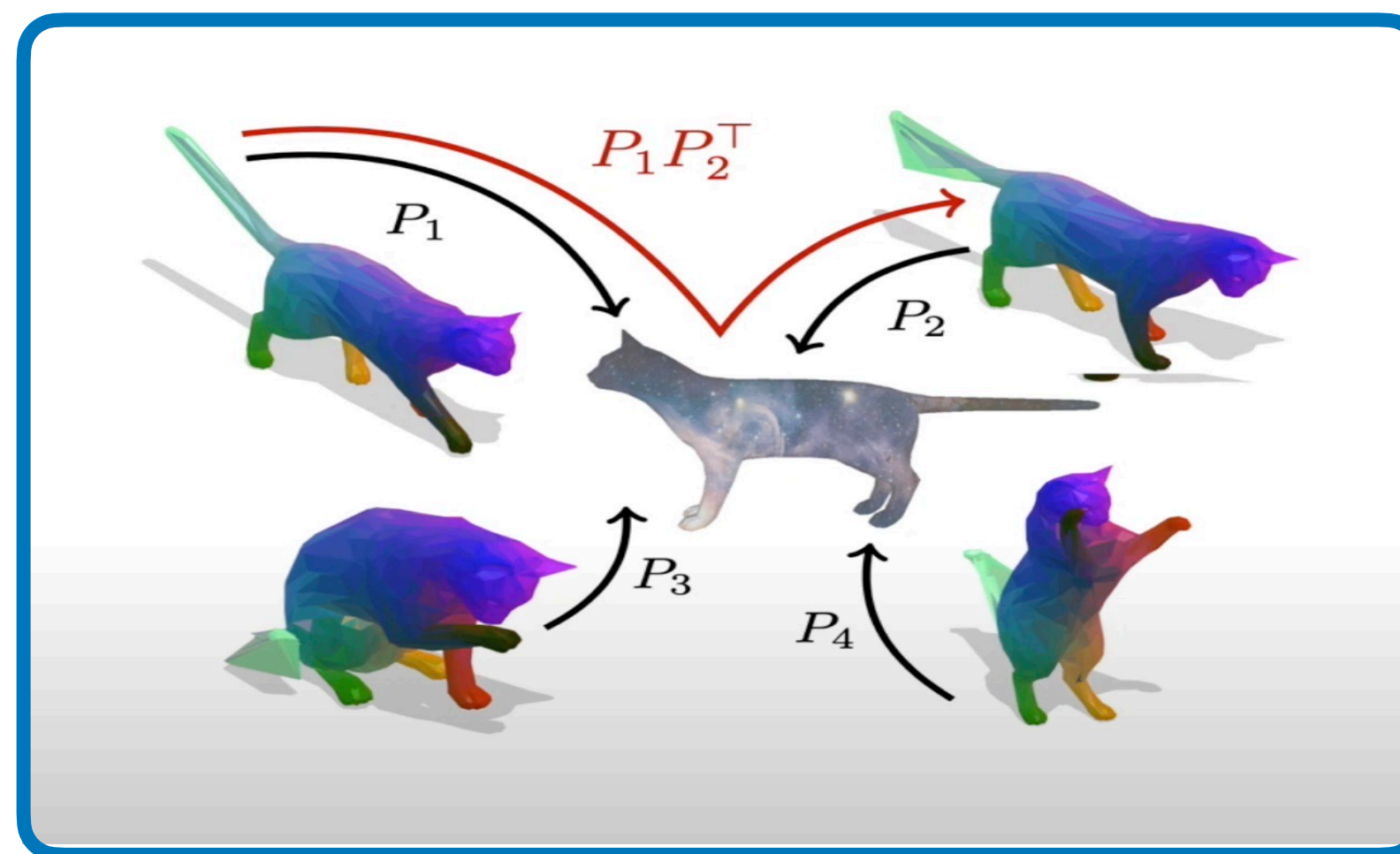
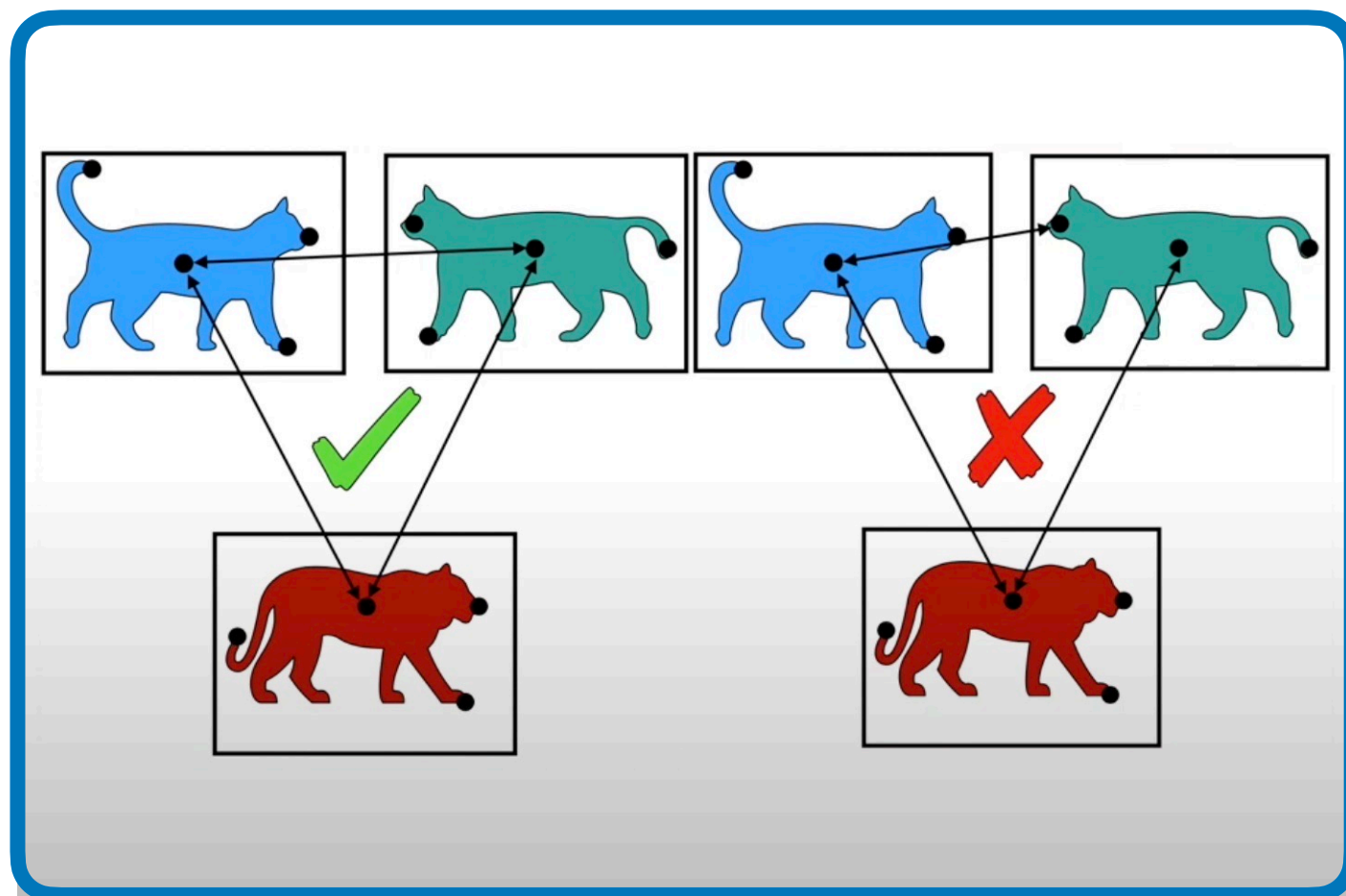
Multi-shape Matching

Input

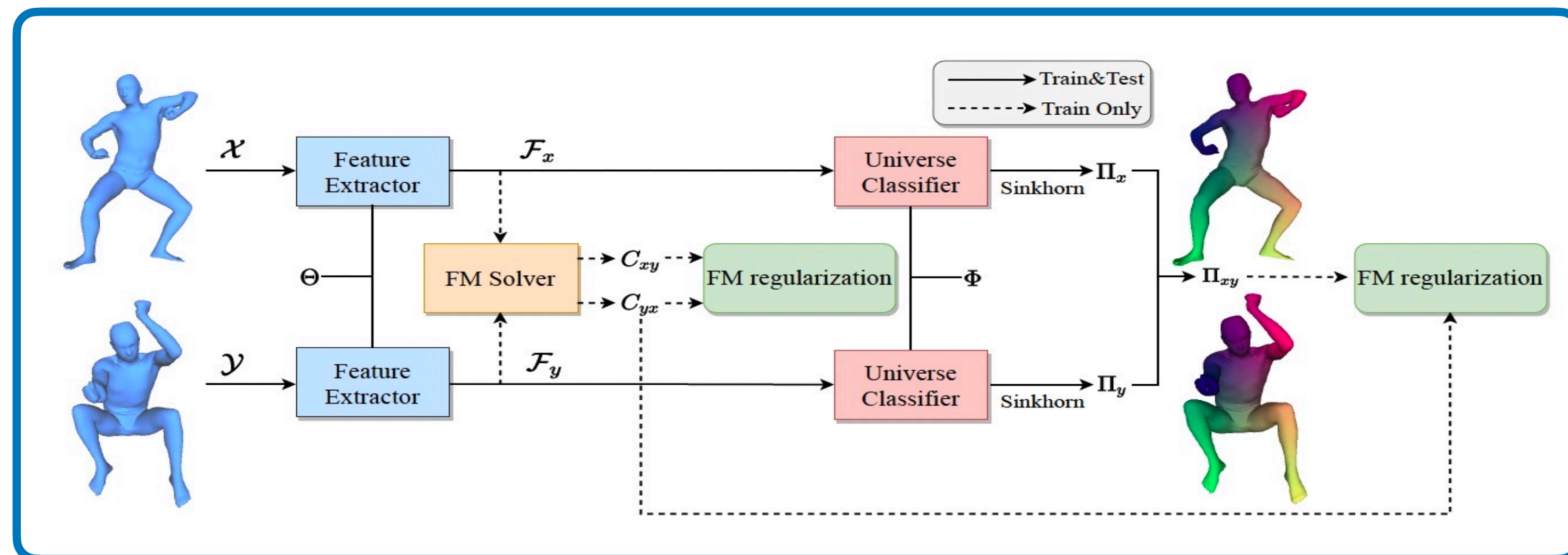
Output



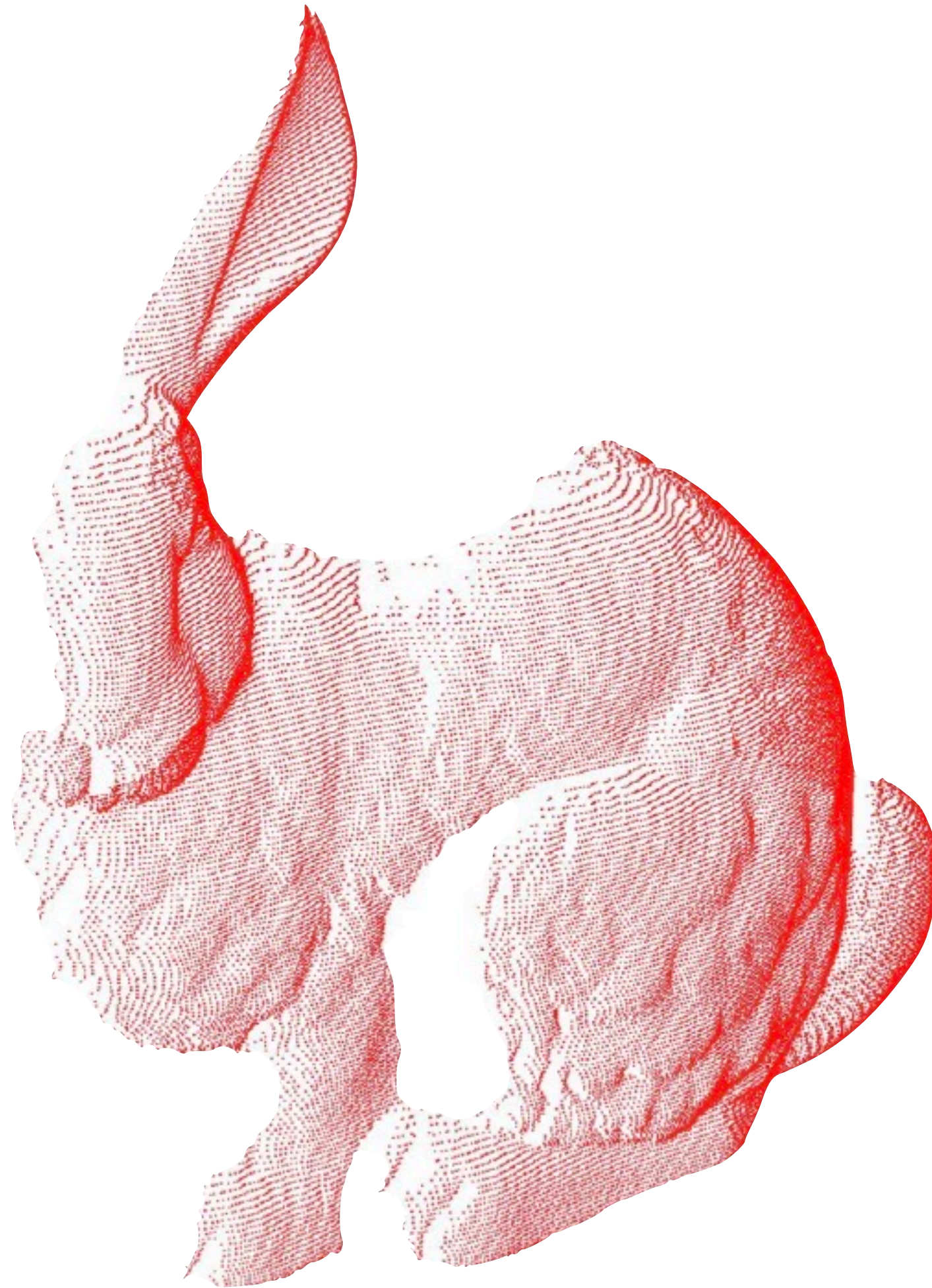
Our Method



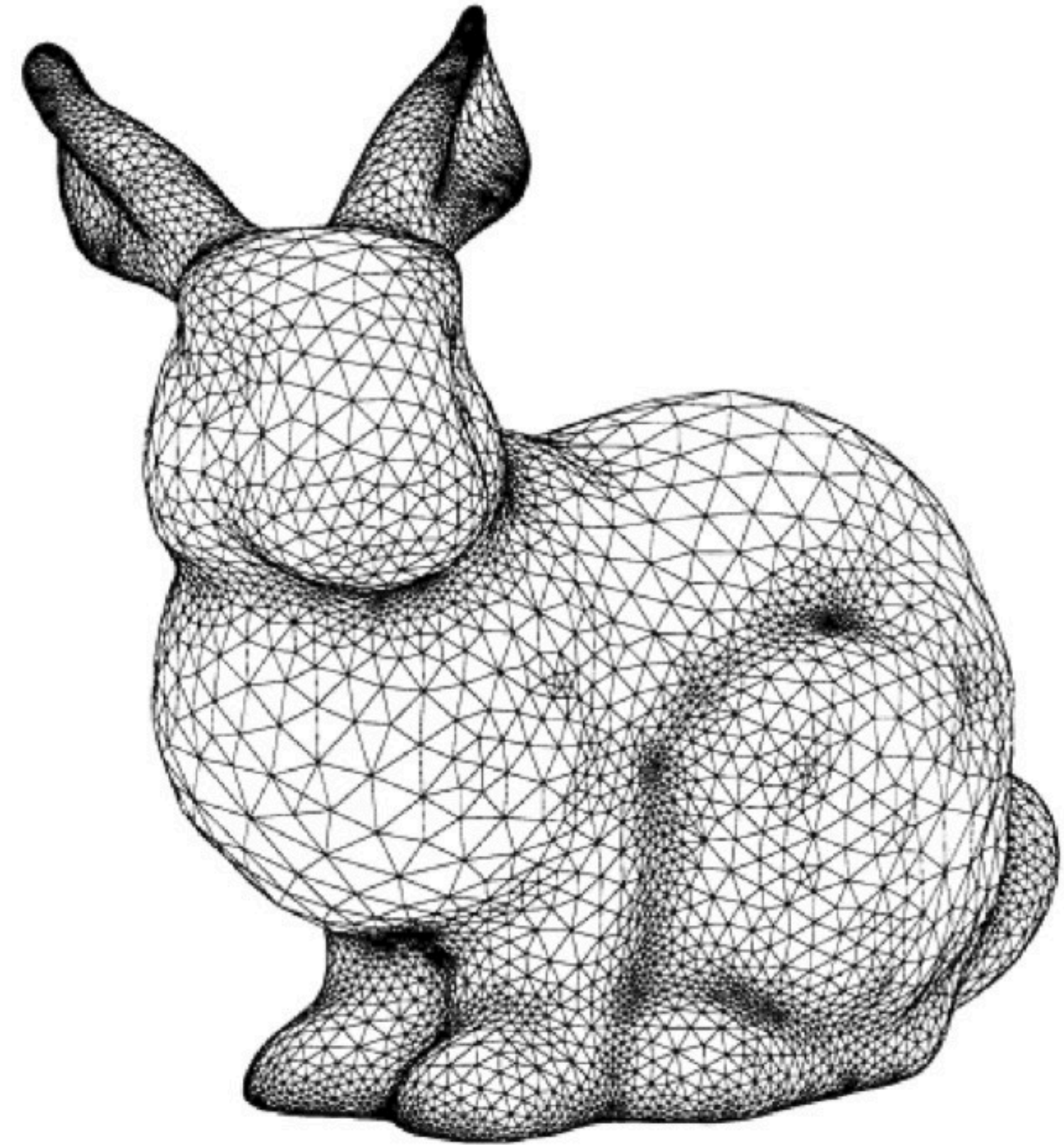
- Unsupervised
- Cyclic Consistent
- No Template
- Partial multi-matching



What is a shape?



Point Cloud

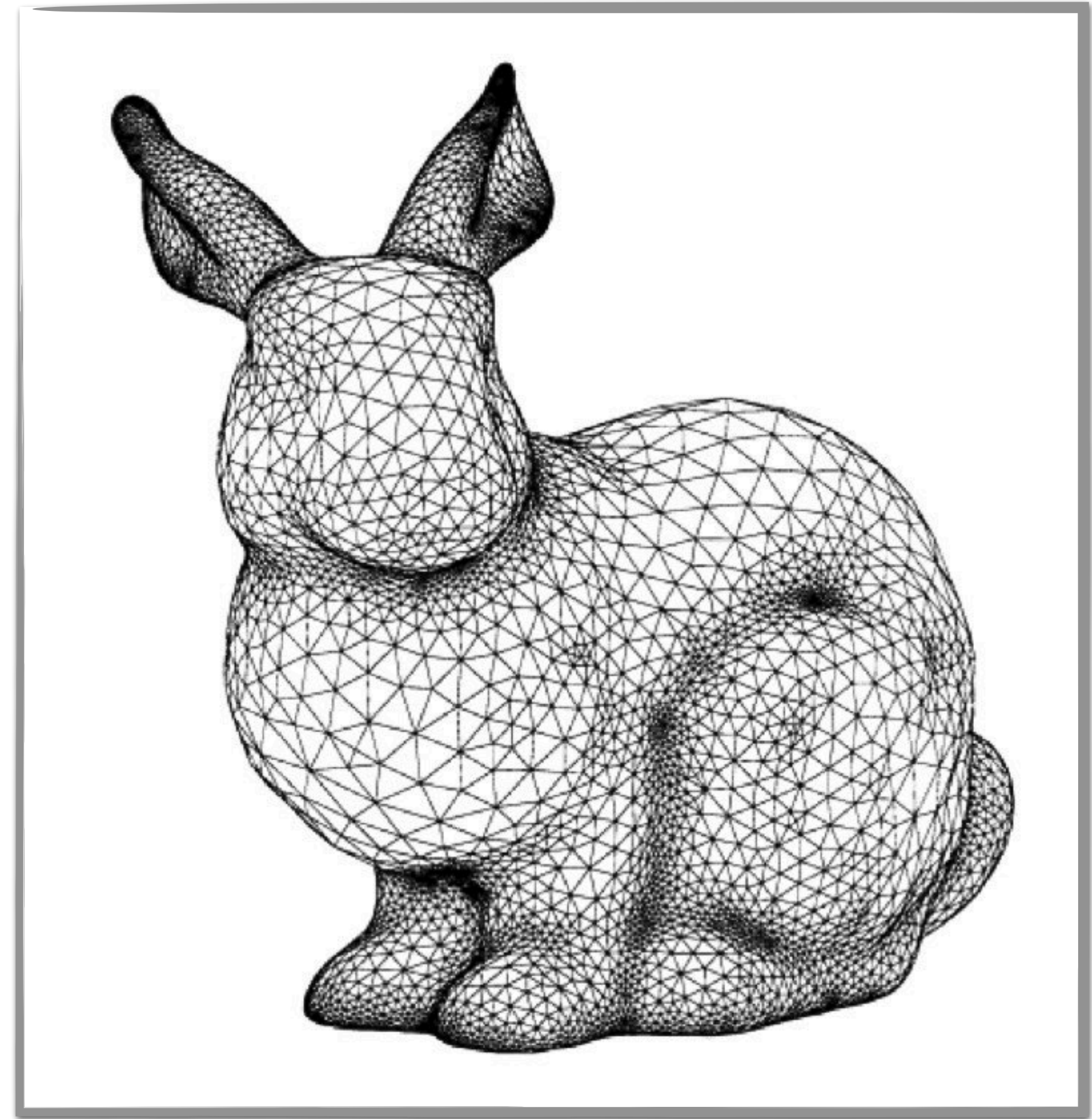


Triangle Mesh

What is a shape?

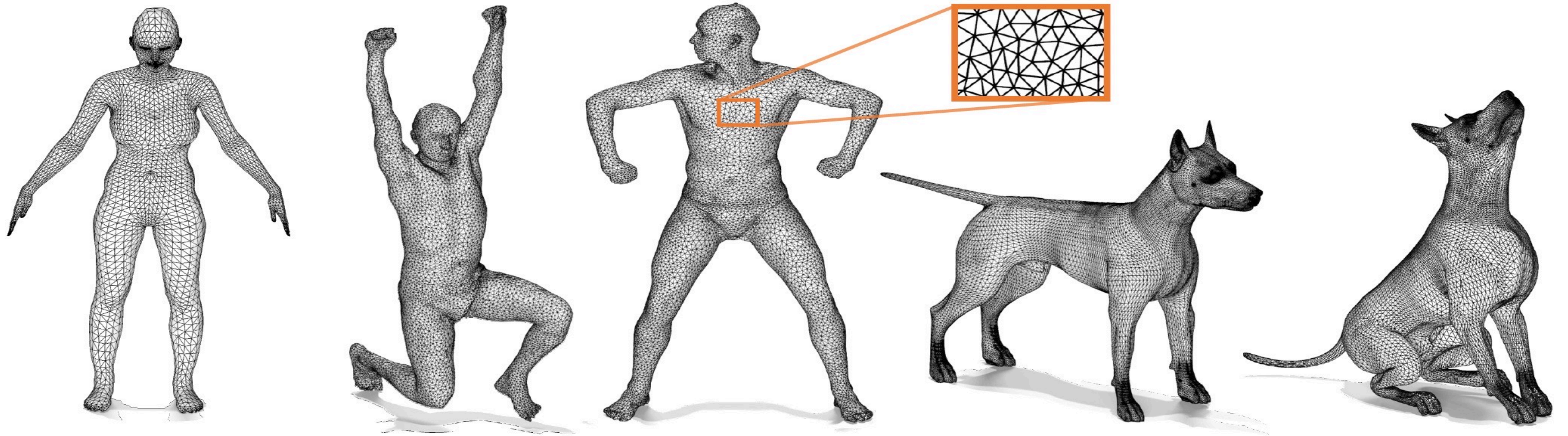


Point Cloud



Triangle Mesh

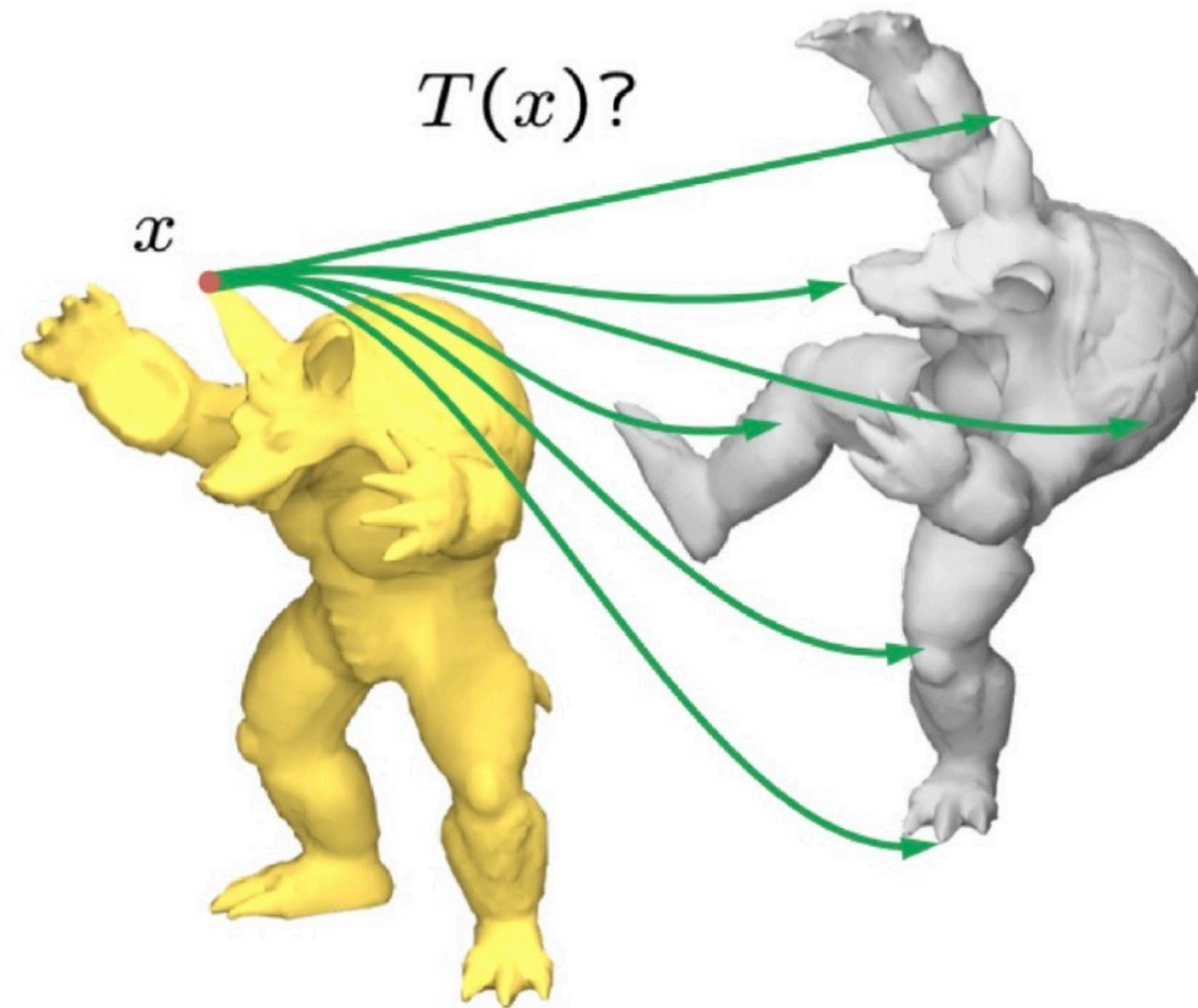
Typical Shapes & Complexity



5k – 200k triangles

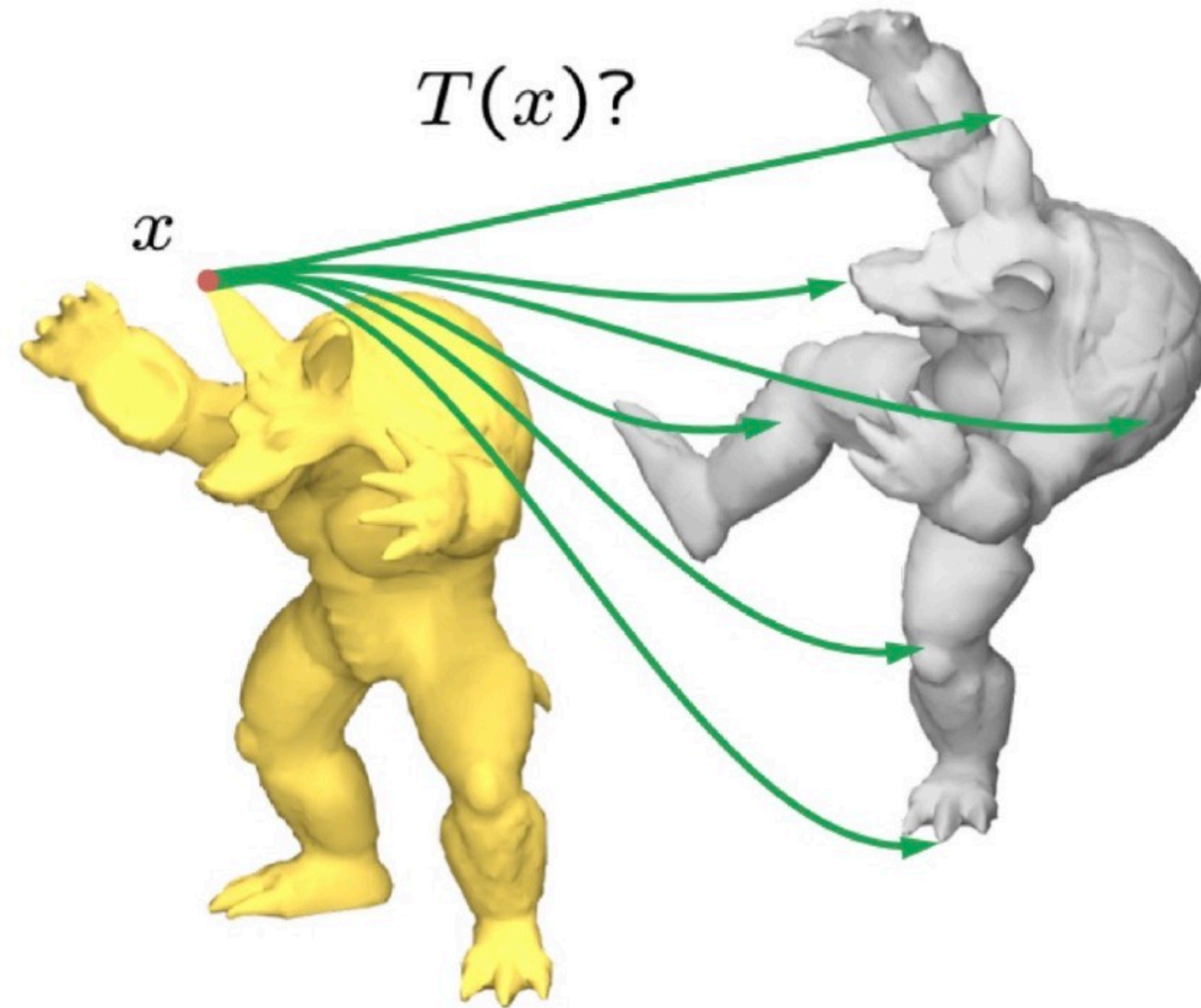
Shapes from the FAUST, SCAPE, and TOSCA datasets

Non-Rigid Shape Matching



No Compact Representation to optimize

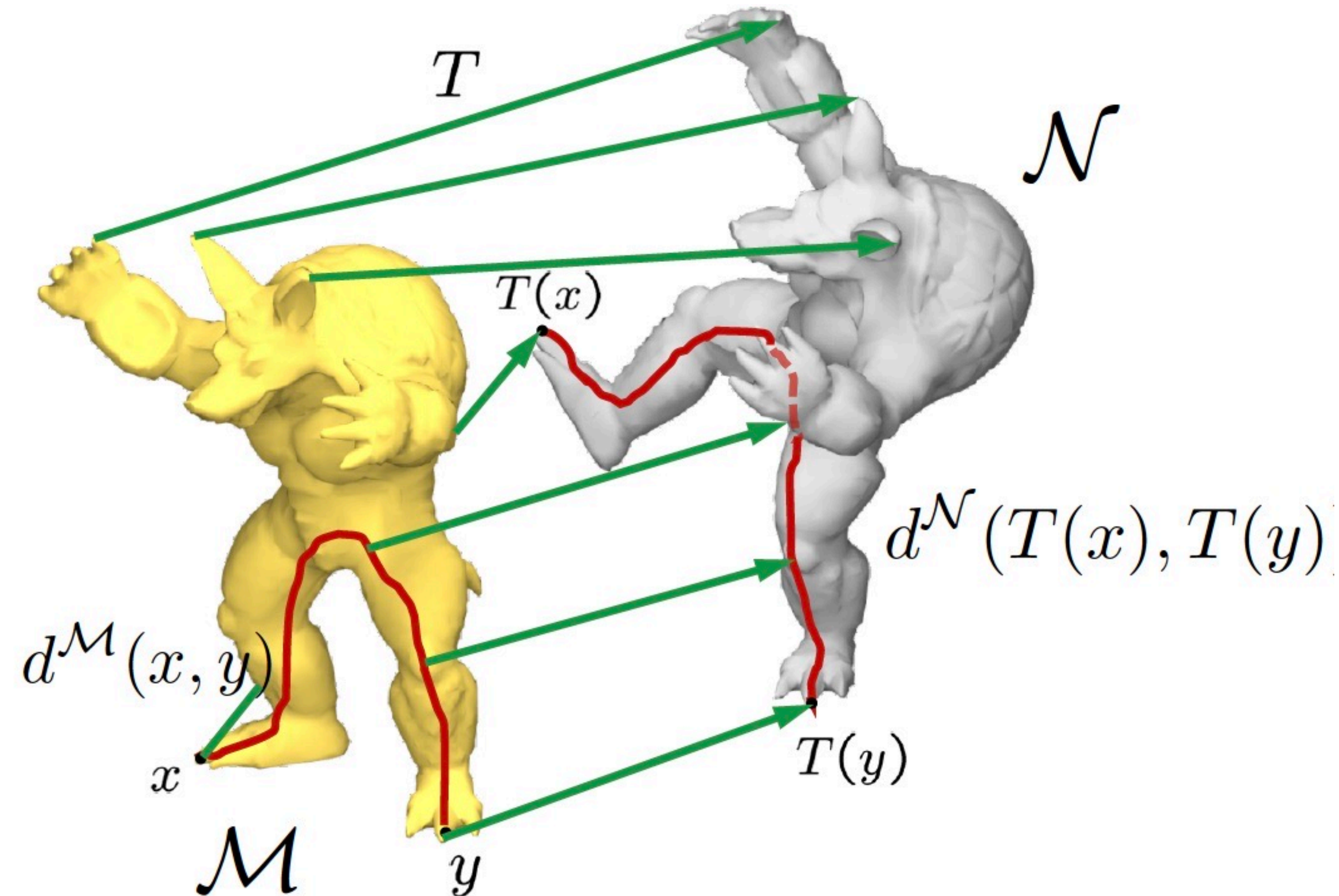
Non-Rigid Shape Matching



Main Questions

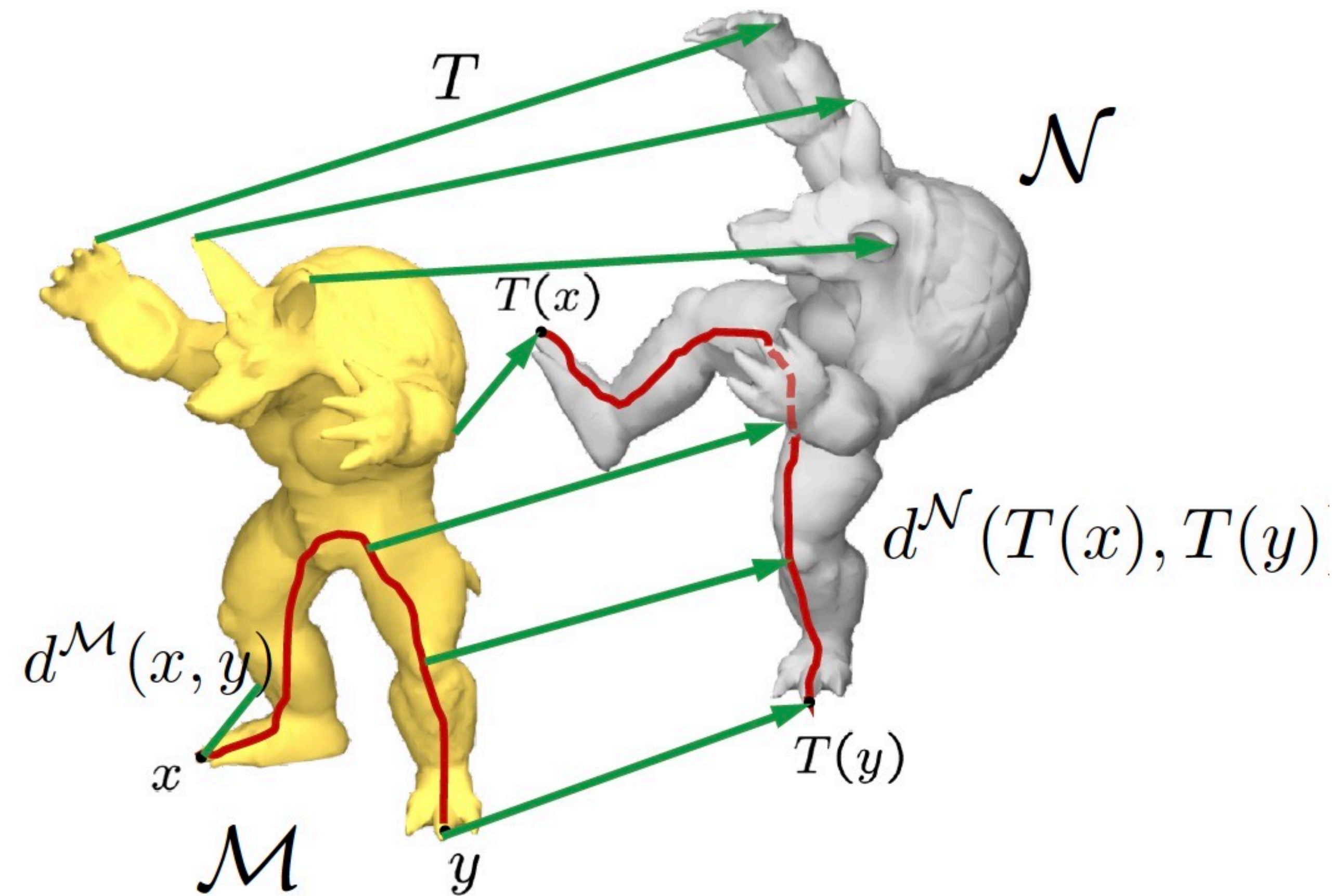
- What does it mean for the correspondence to be good?
- How to compute it efficiently?

Isometric Shape Matching



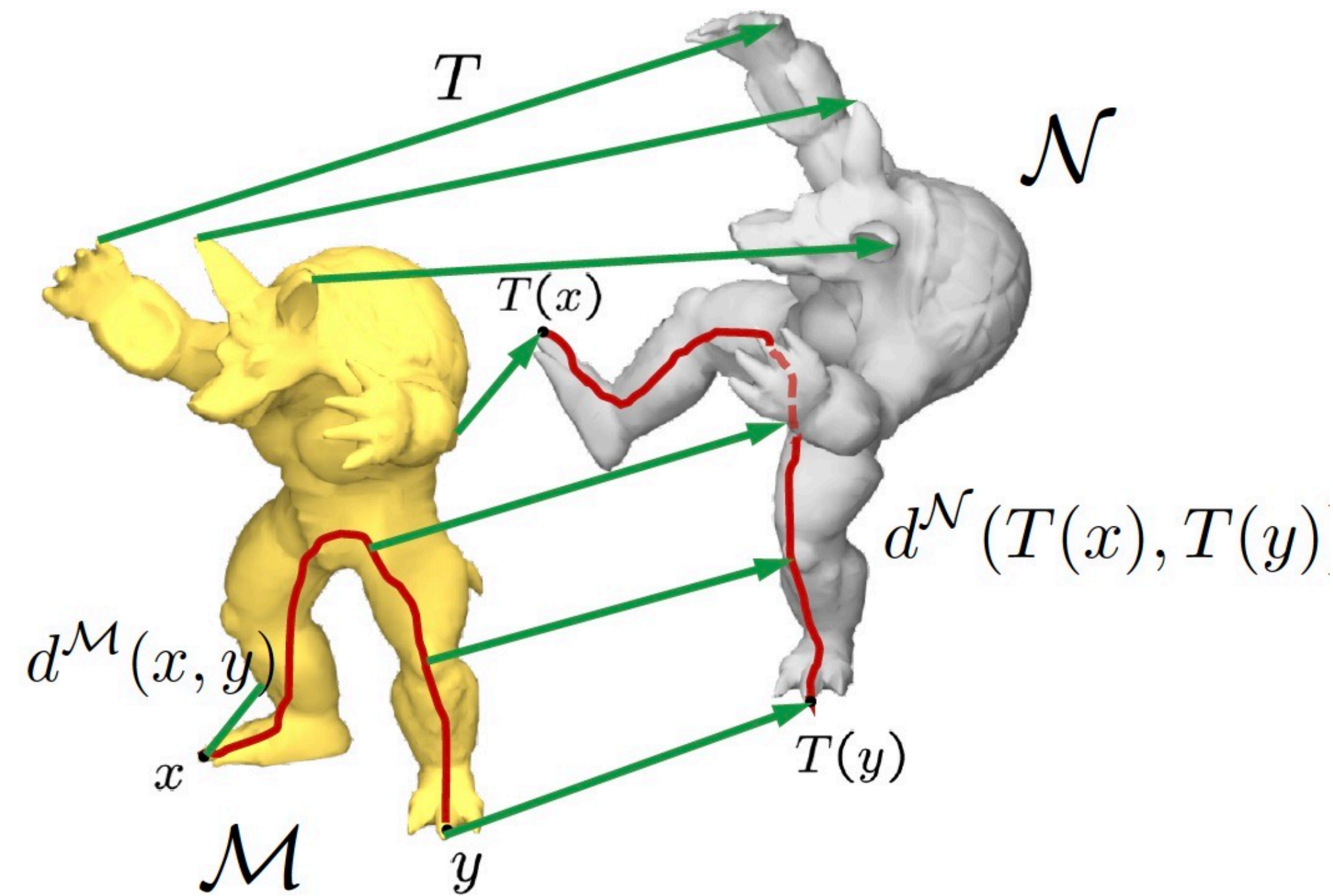
Good maps must preserve geodesic distances

Isometric Shape Matching



$$T_{\text{opt}} = \arg \min_T \sum_{x, y} \|d^{\mathcal{M}}(x, y) - d^{\mathcal{N}}(T(x), T(y))\|$$

Isometric Shape Matching - Challenges

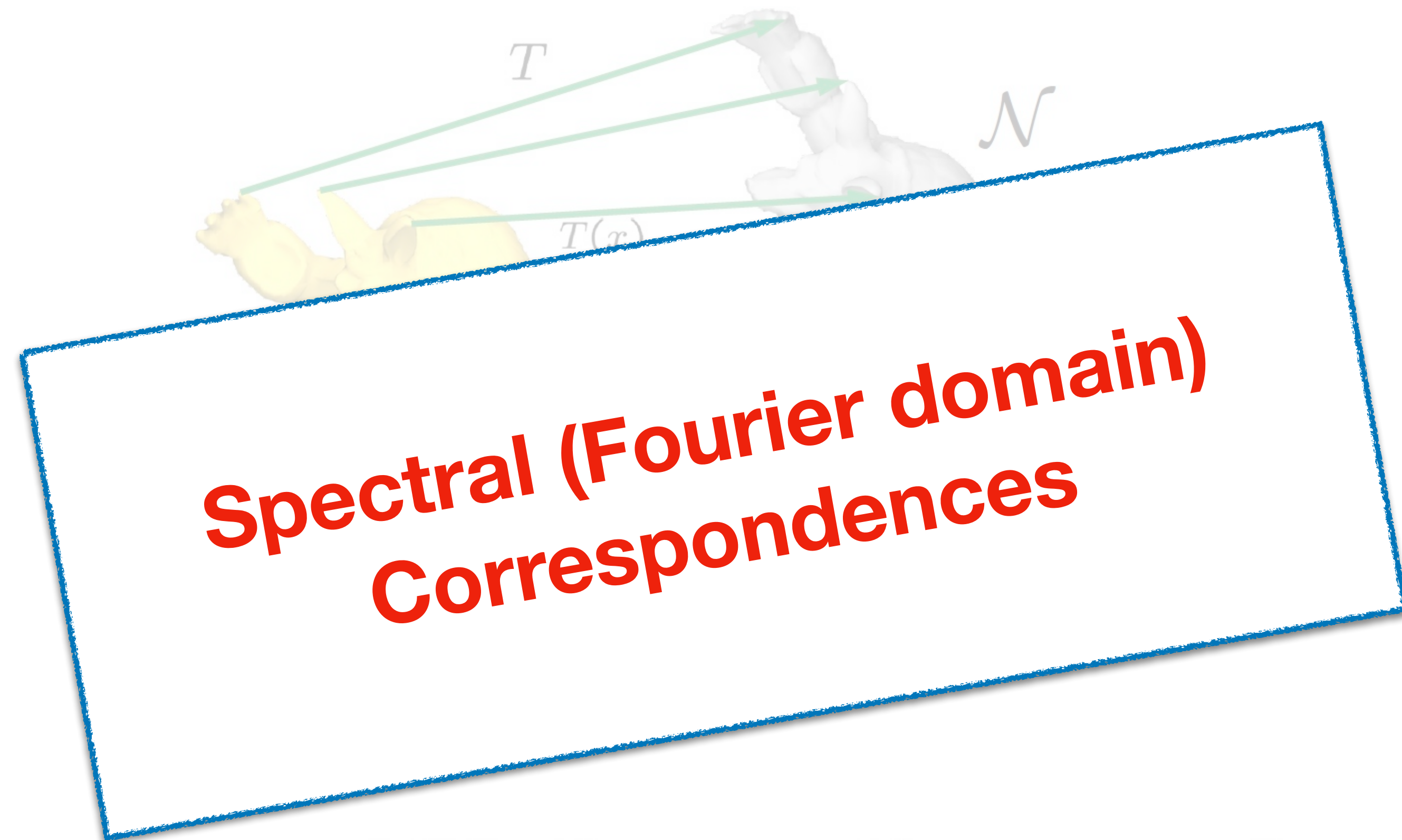


$$T_{\text{opt}} = \arg \min_T \sum_{x, y} \|d^{\mathcal{M}}(x, y) - d^{\mathcal{N}}(T(x), T(y))\|$$

Problem

The space of possible solutions is non linear and non convex

Isometric Shape Matching - Challenges



$$T_{\text{opt}} = \arg \min_T \sum_{x,y} \|d^{\mathcal{M}}(x,y) - d^{\mathcal{N}}(T(x),T(y))\|$$

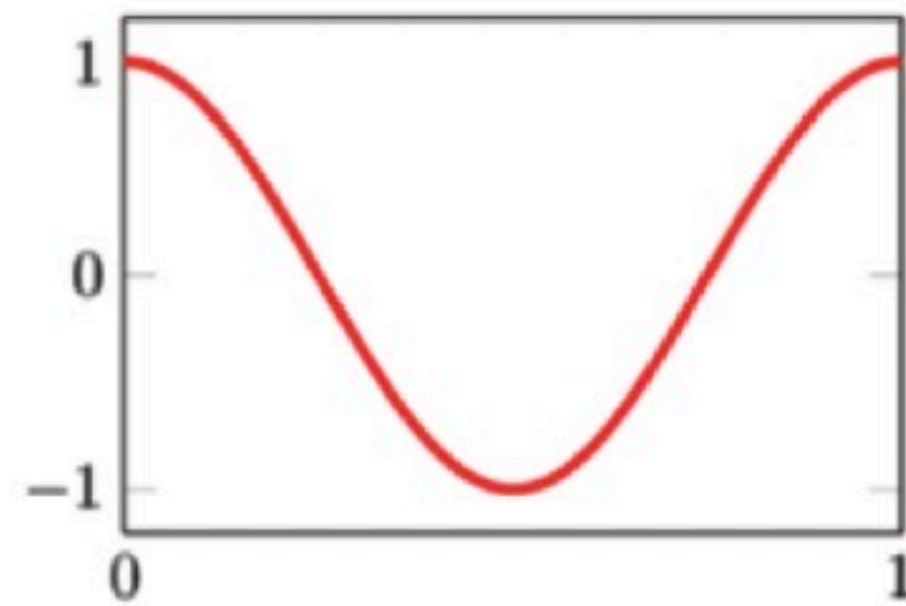
Problem

The space of possible solutions is non linear and non convex

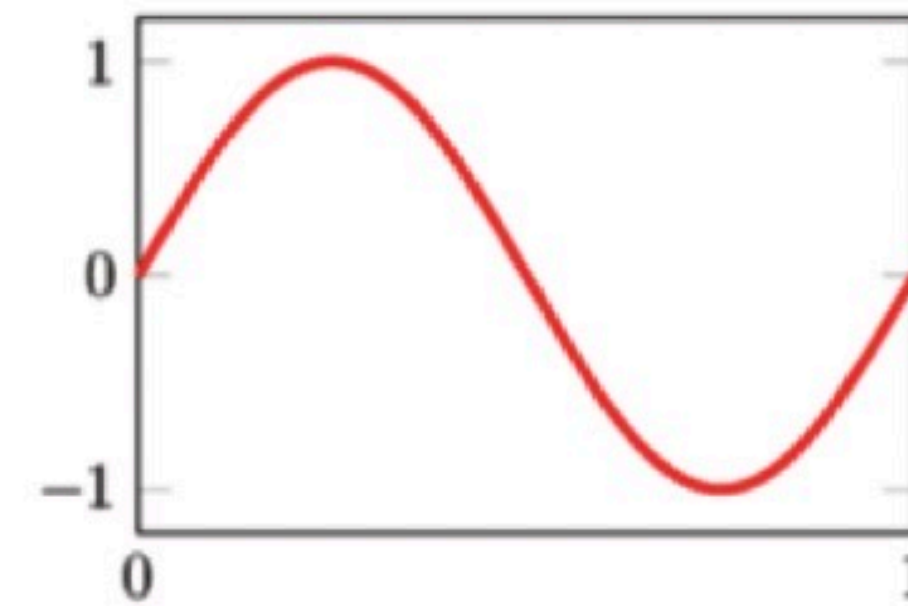
Fourier Bases



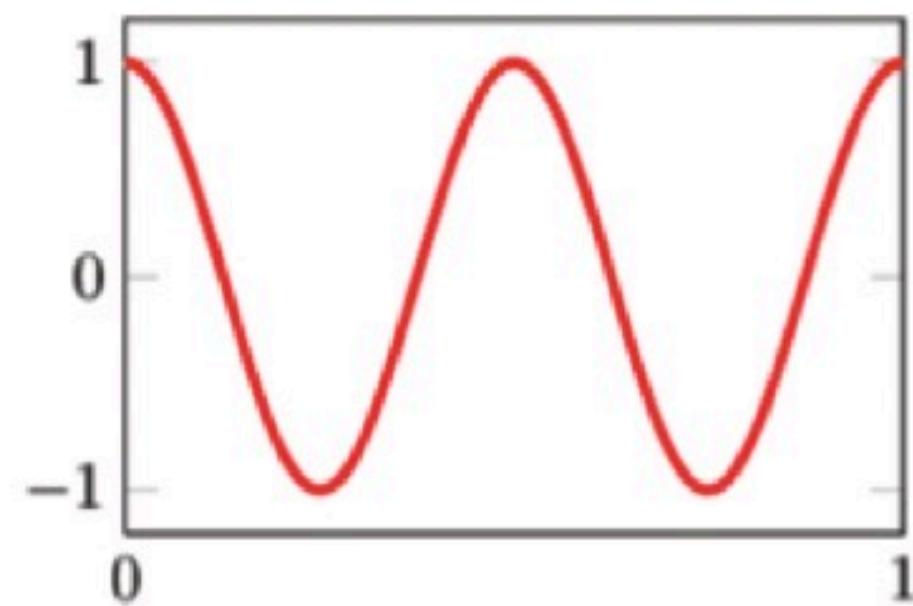
(a) constant term



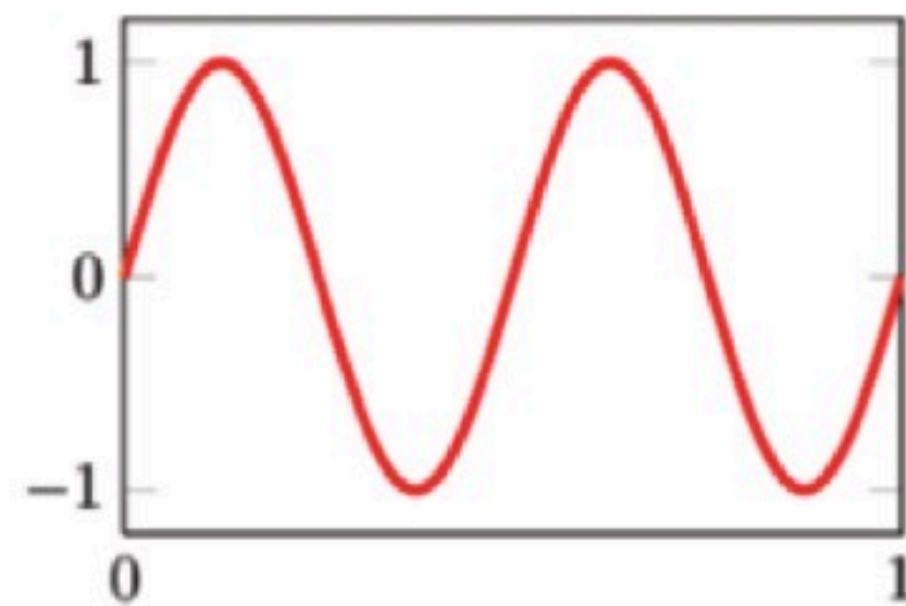
(b) $\cos(2\pi t)$



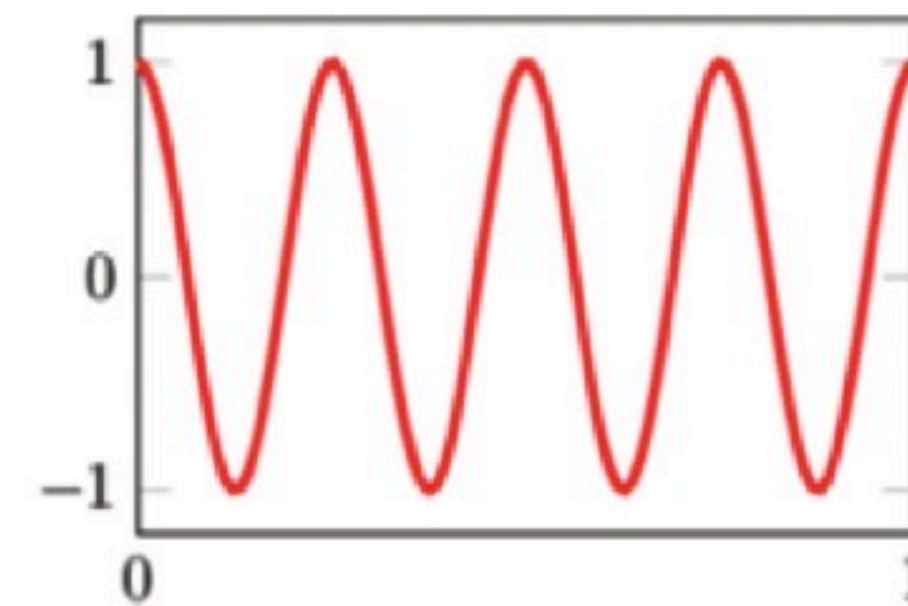
(c) $\sin(2\pi t)$



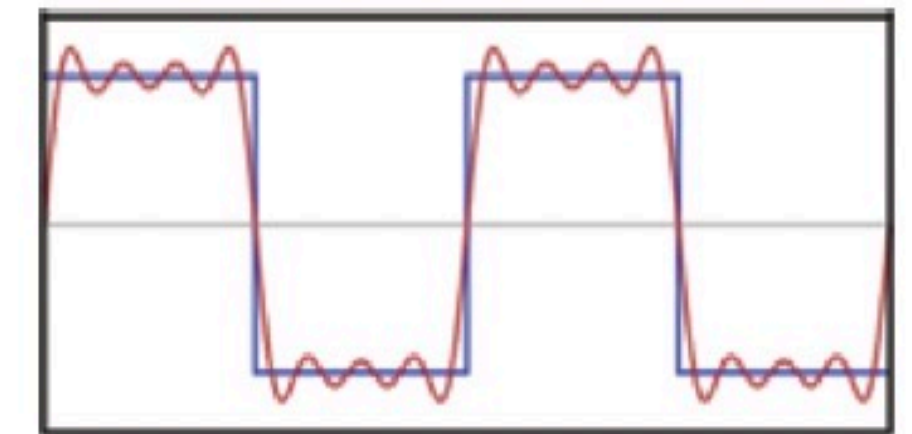
(d) $\cos(4\pi t)$



(e) $\sin(4\pi t)$



(f) $\cos(6\pi t)$

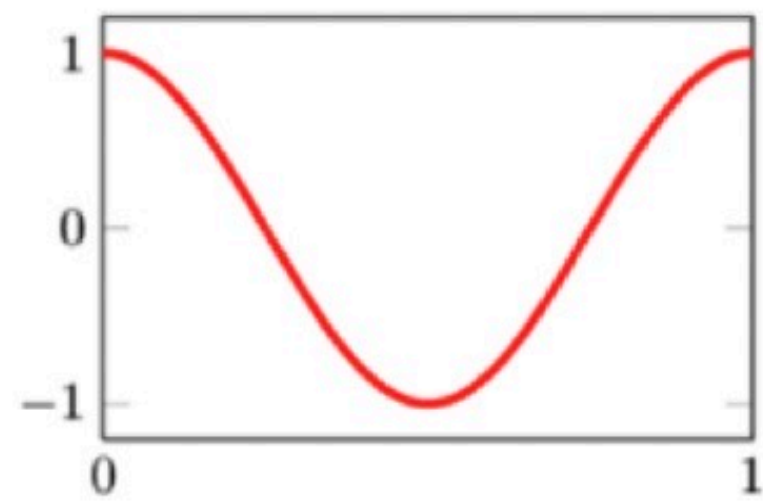


$$f(t) = a_0 + \sum_{k=1}^{\infty} (a_k \sin(2\pi kt) + b_k \cos(2\pi kt))$$

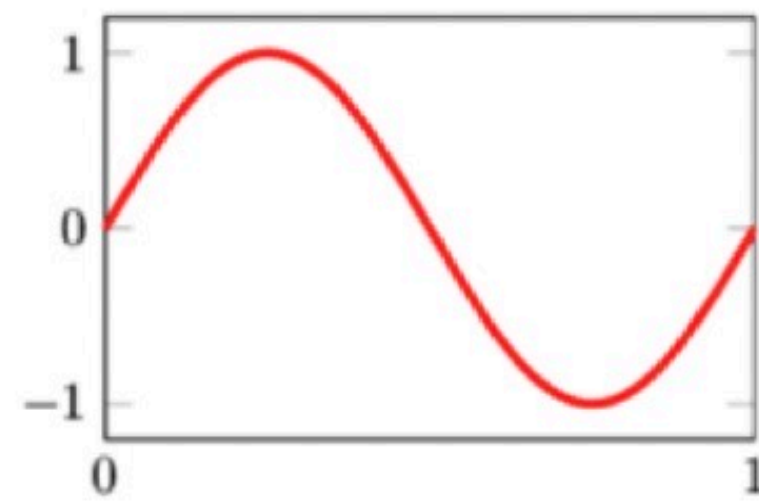
Functional Bases



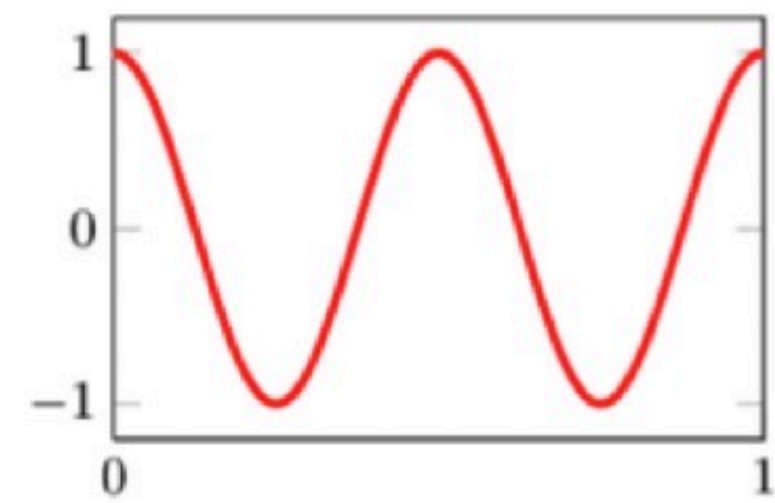
(a) constant term



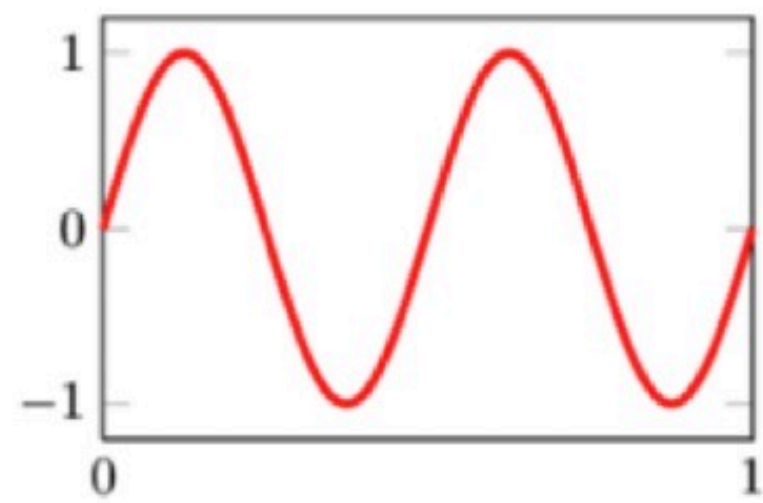
(b) $\cos(2\pi t)$



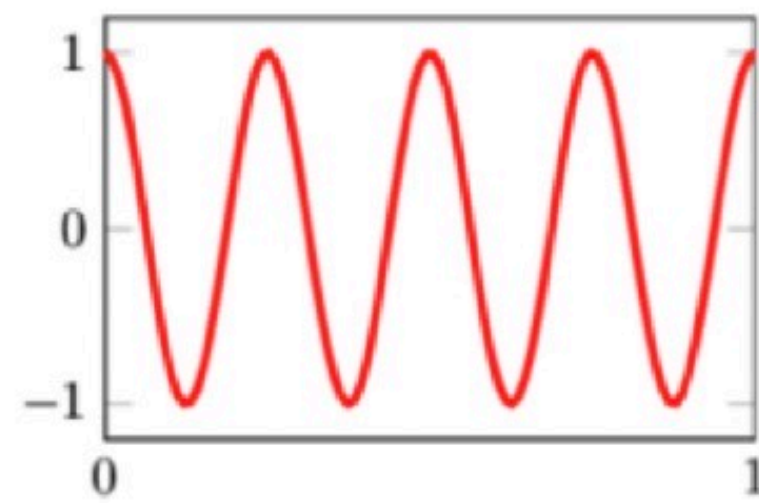
(c) $\sin(2\pi t)$



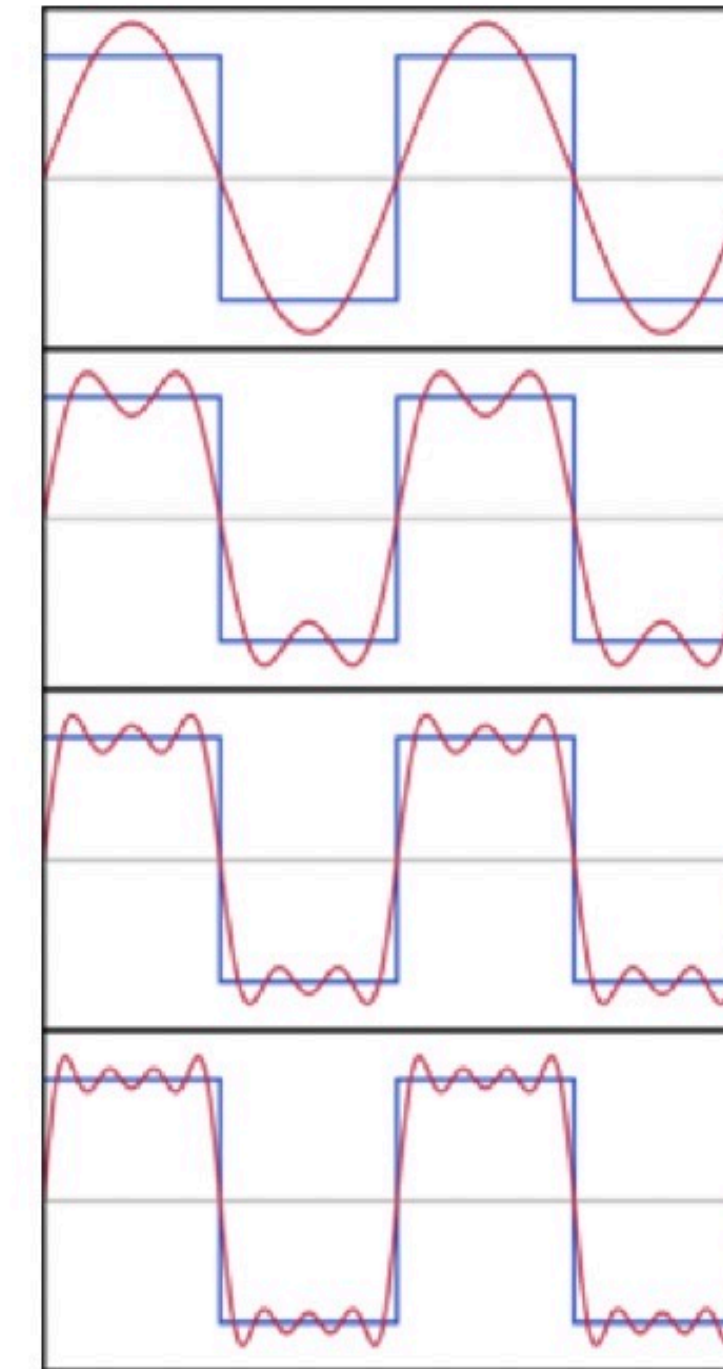
(d) $\cos(4\pi t)$



(e) $\sin(4\pi t)$



(f) $\cos(6\pi t)$



$$\sum_{k=1}^2 a_k \varphi_k(t)$$

$$\sum_{k=1}^3 a_k \varphi_k(t)$$

$$\sum_{k=1}^4 a_k \varphi_k(t)$$

$$\sum_{k=1}^5 a_k \varphi_k(t)$$

$$f(t) = \sum_{k=1}^N a_k \varphi_k(t)$$

$$a_k = \langle f, \varphi_k \rangle = \int_0^1 f(t) \varphi_k(t) dt$$

Laplace -Beltrami eigenfunctions

Eigenfunctions of the *Laplacian* matrix

$$L\phi_i = \lambda_i \phi_i$$

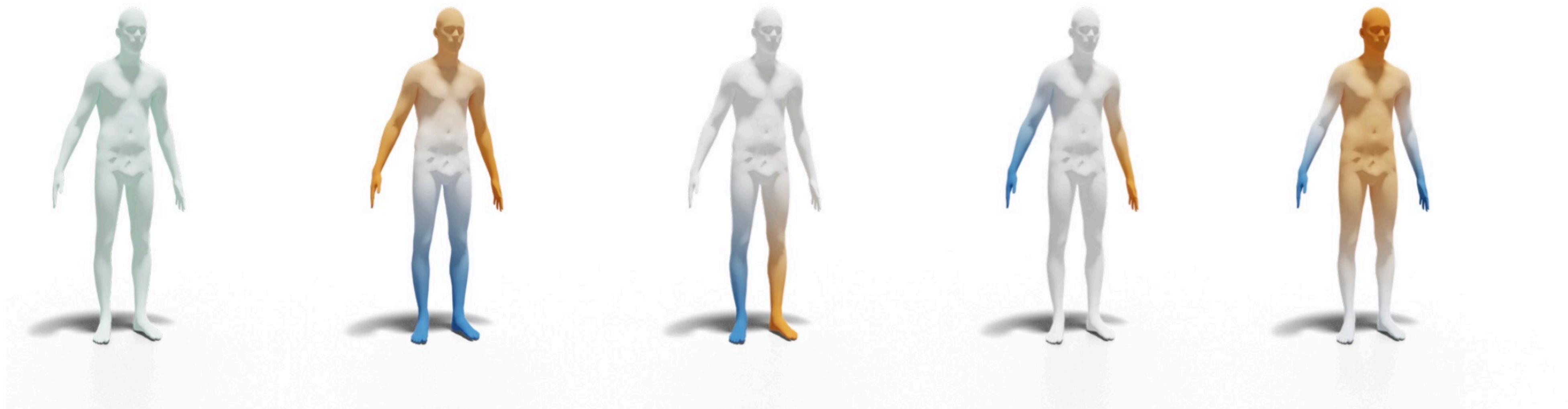
$$\lambda_0 = 0$$

$$\lambda_1 = 2.39$$

$$\lambda_2 = 3.11$$

$$\lambda_3 = 5.00$$

$$\lambda_4 = 7.31$$

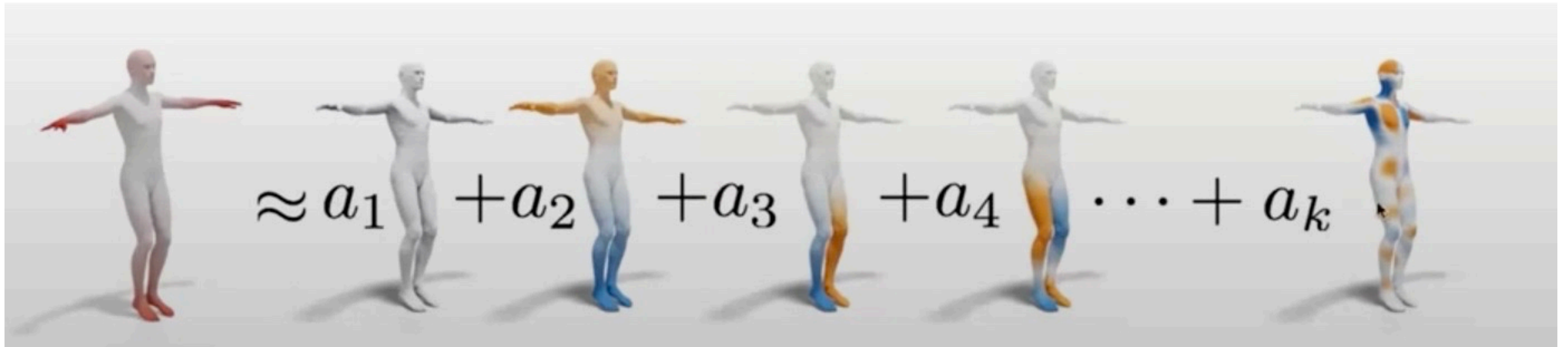


“Laplace-Beltrami Eigenfunctions Towards an algorithm that “understands” geometry,” Bruno Levy
IEEE International Conference on Shape Modeling and Applications 2006

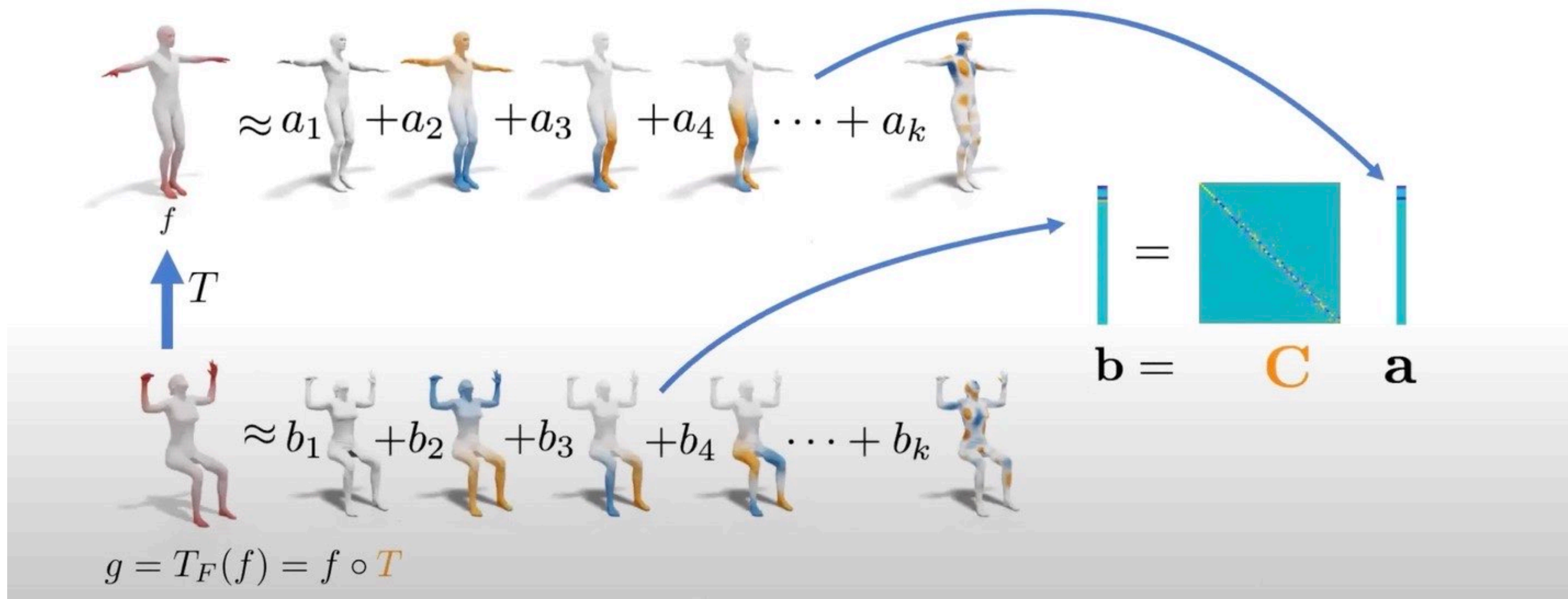
Signal Processing on Surfaces

$$f = \sum_{i=0}^{\infty} a_i \phi_i$$

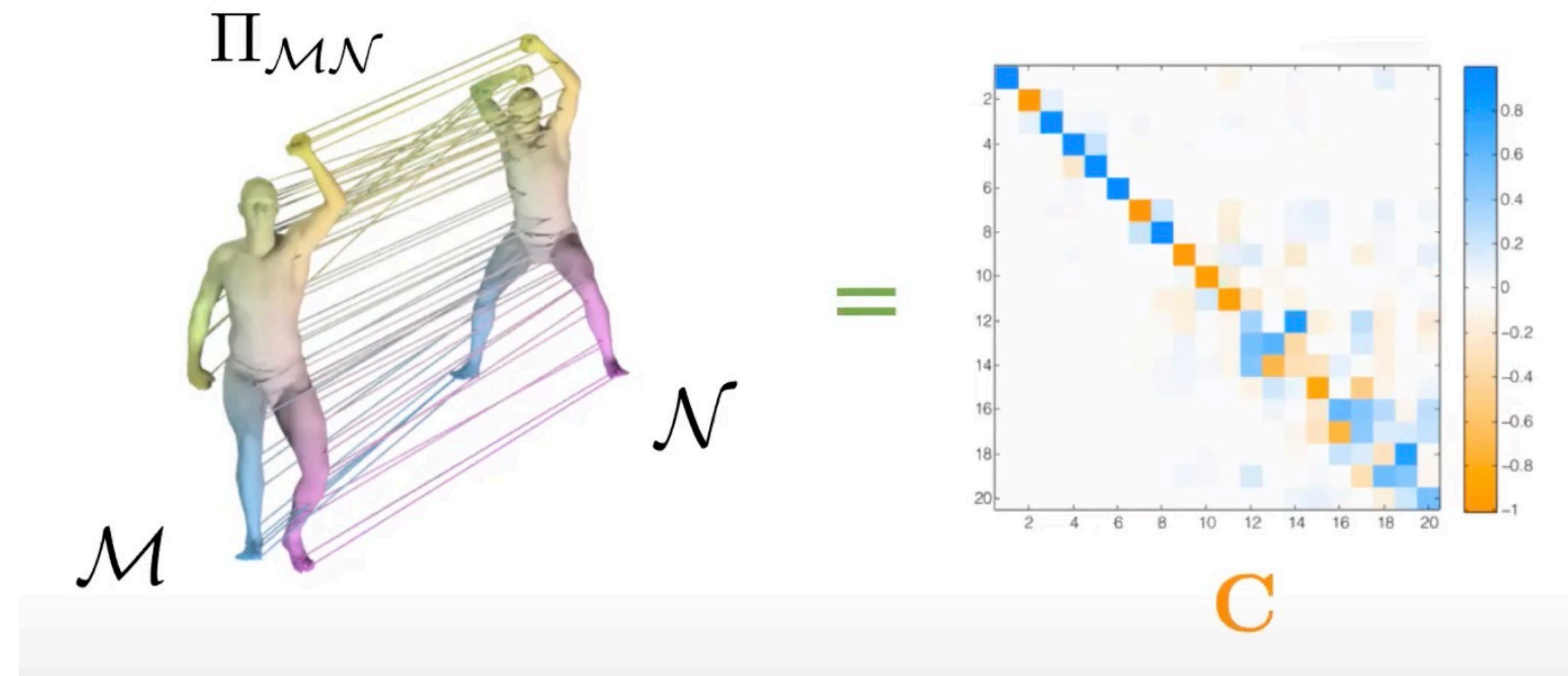
$$a_i = \int_{\mathcal{M}} f(x) \phi_i(x) d\mu(x)$$



Functional Maps



Functional Maps Representation

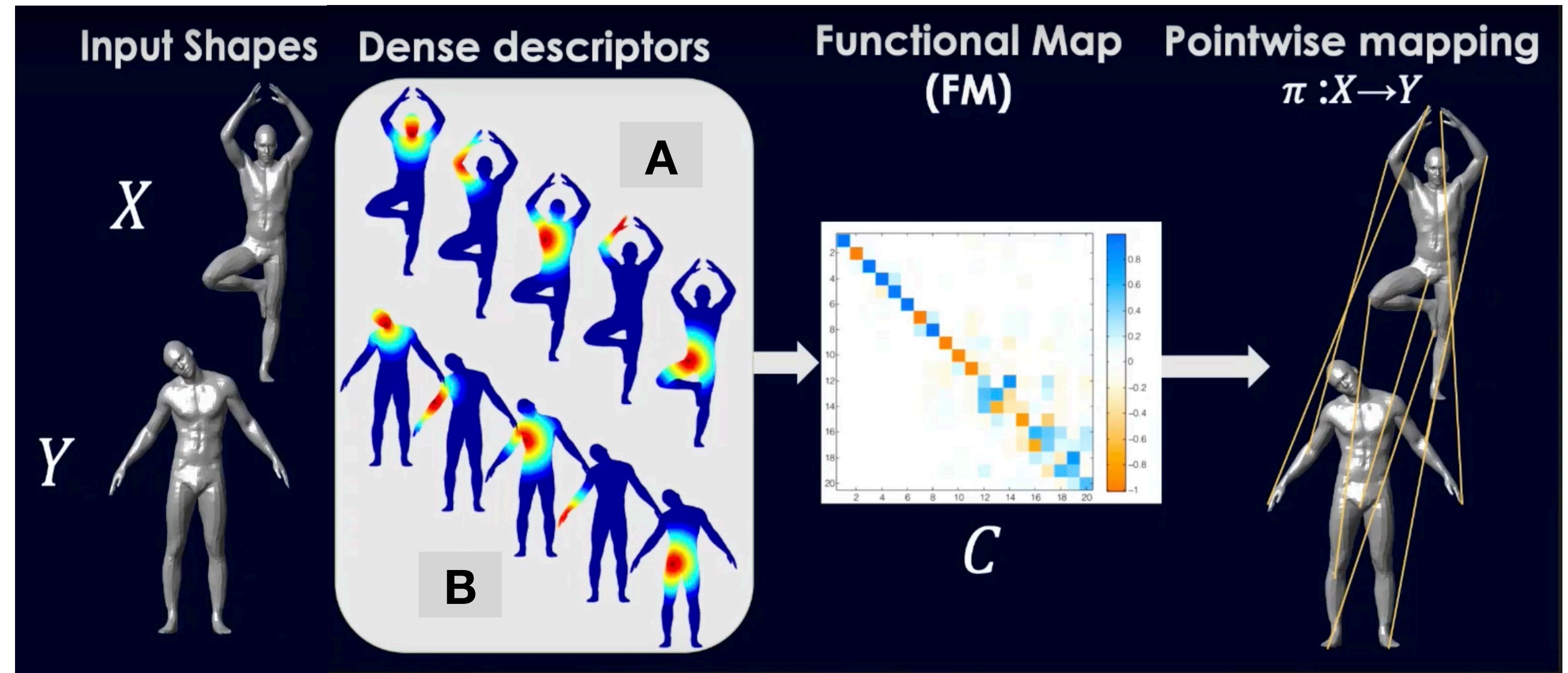


Main advantages

- $\mathbf{C}$ is much smaller than $\Pi_{\mathcal{M}\mathcal{N}}$
- Natural Constraints are easy to express

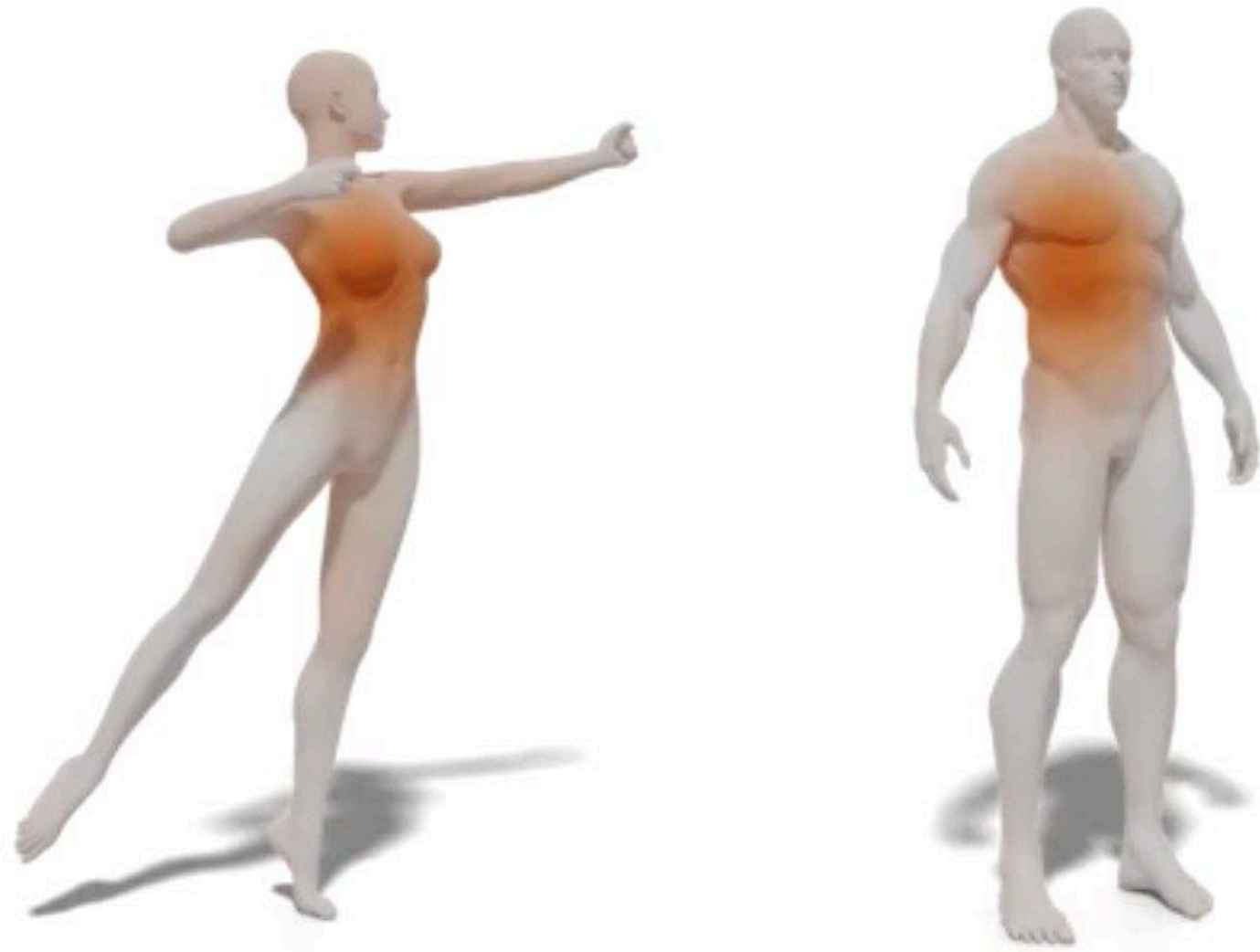
Functional Map Pipeline

1. Compute first **k basis functions** of two shapes.
2. Write descriptor functions in basis : **A** , **B**
3. Solve $C_{\text{opt}} = \arg \min_C \|C\mathbf{A} - \mathbf{B}\|^2$
4. Convert C_{opt} back to point -wise map.

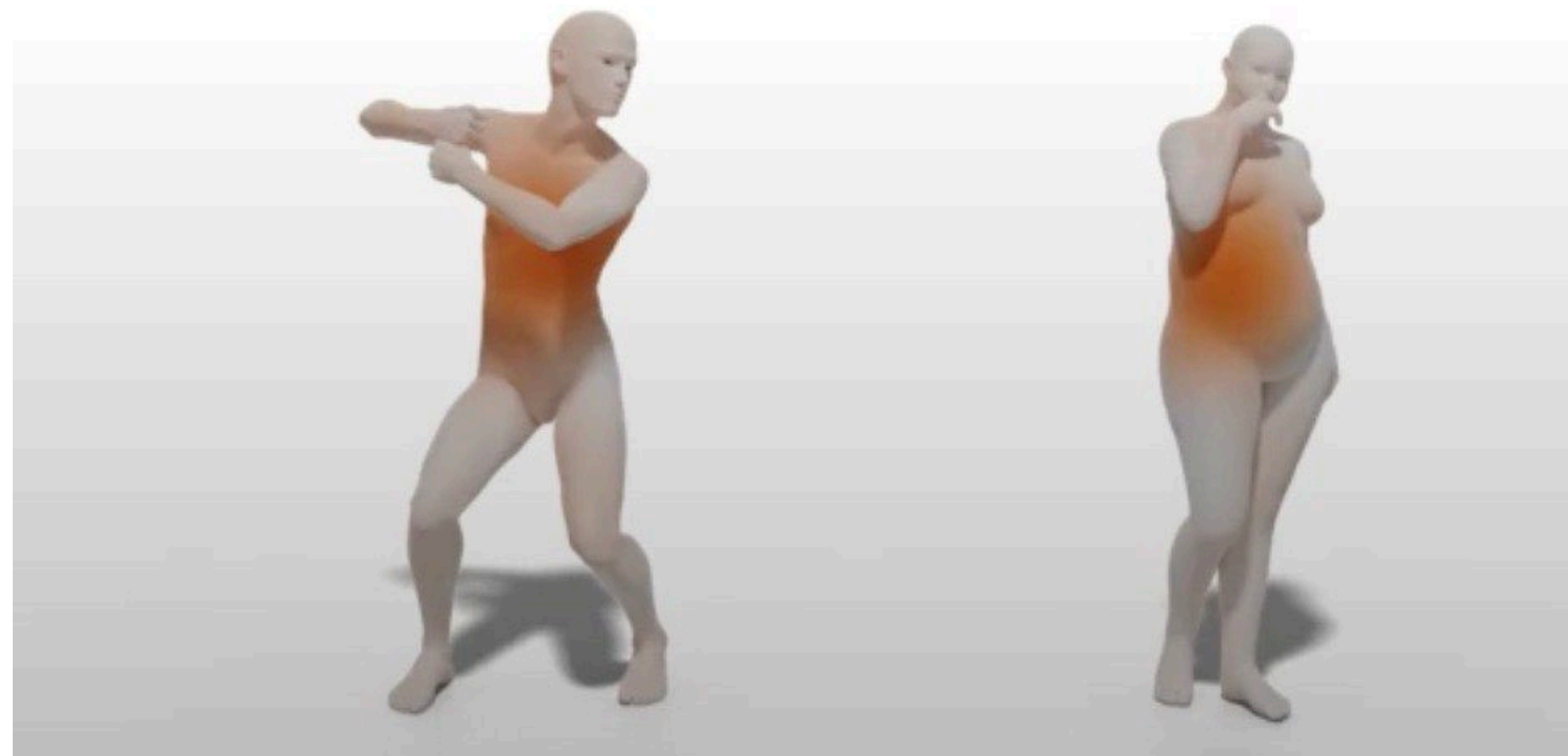


“Functional Maps: A Flexible Representation of Maps Between Shapes”, Ovsjanikov, M., Ben-Chen, M., Solomon, J., Butscher, A., Guibas, SIGGRAPH 2012

Descriptor functions

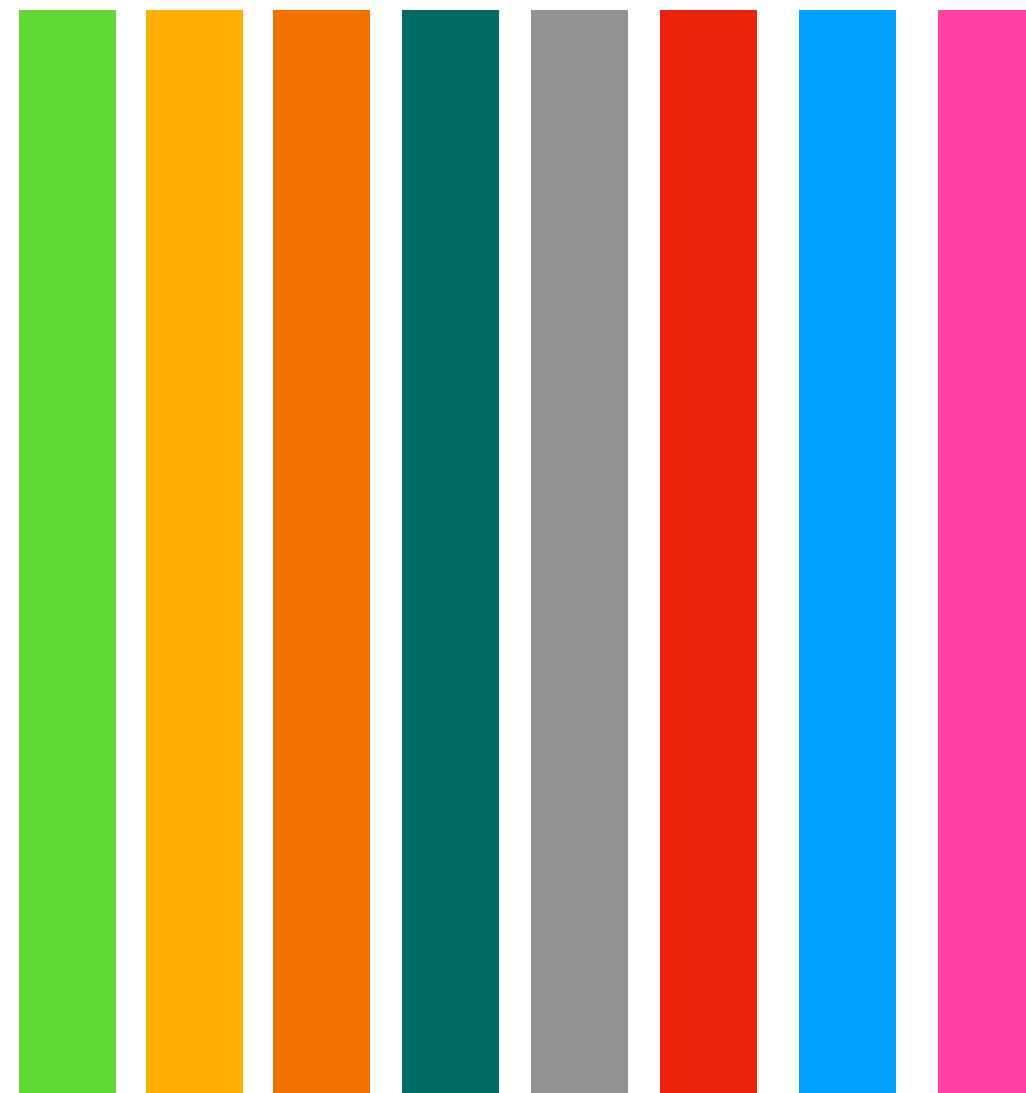


Ideally **Invariant** under non rigid transformation

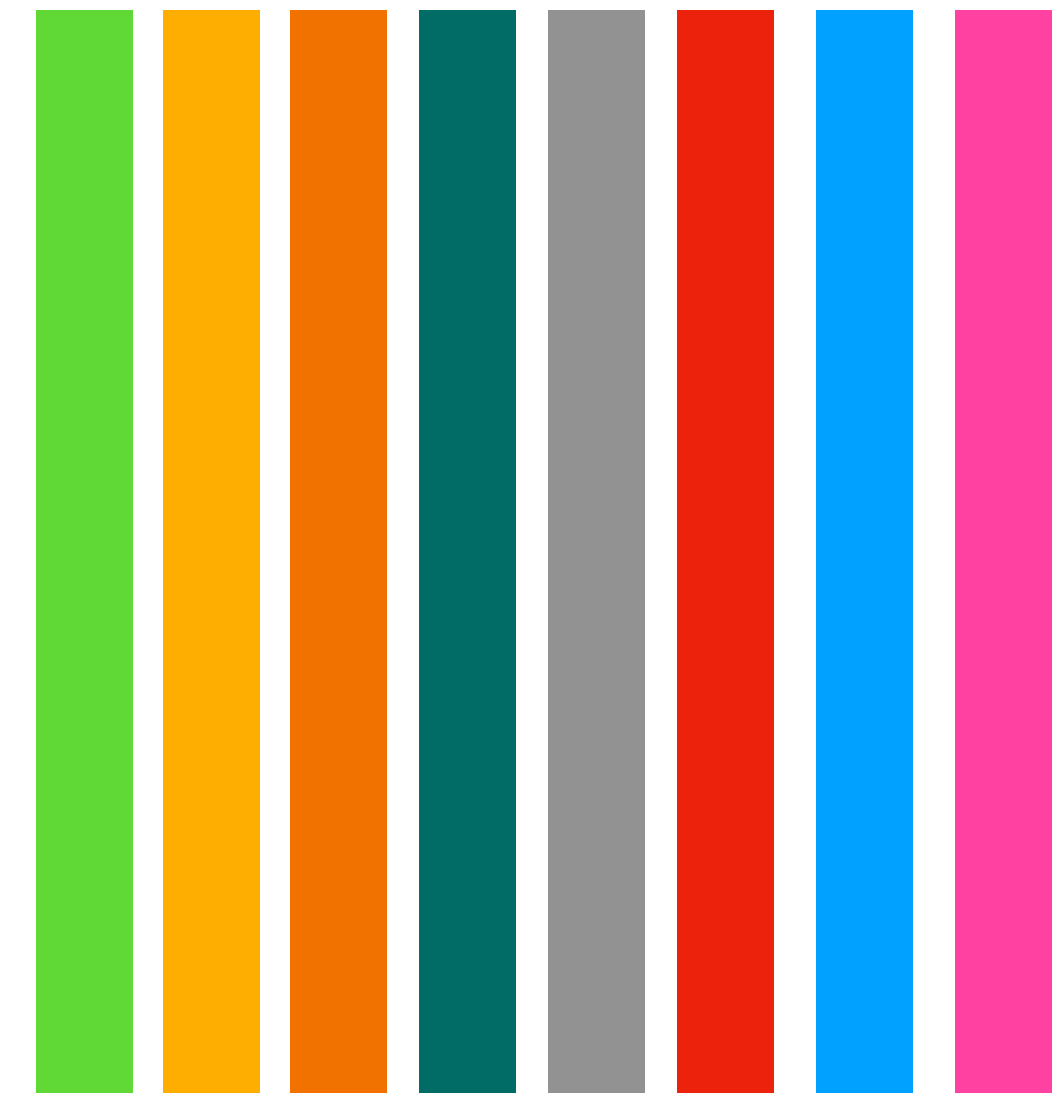


Functional Map Solution

$$C_{\text{opt}} = \arg \min_C \|C\mathbf{A} - \mathbf{B}\|^2$$



$\approx$



$$B_{k \times q} = (\langle \psi_i, g_j \rangle_Y)$$

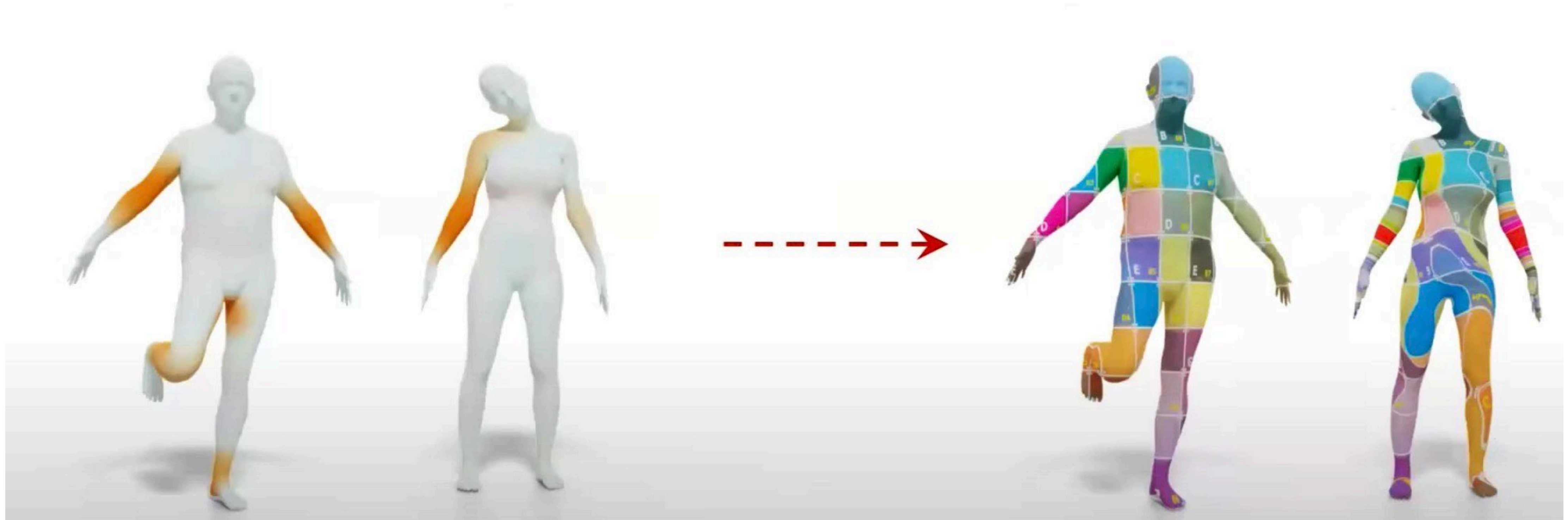
$$C_{k \times k}$$

$$A_{k \times q} = (\langle \phi_i, f_j \rangle_X)$$

A and **B** are Fourier coefficients of corresponding descriptor functions

$$g_i \approx Tf_i$$

What happens if input descriptors are **bad** ?

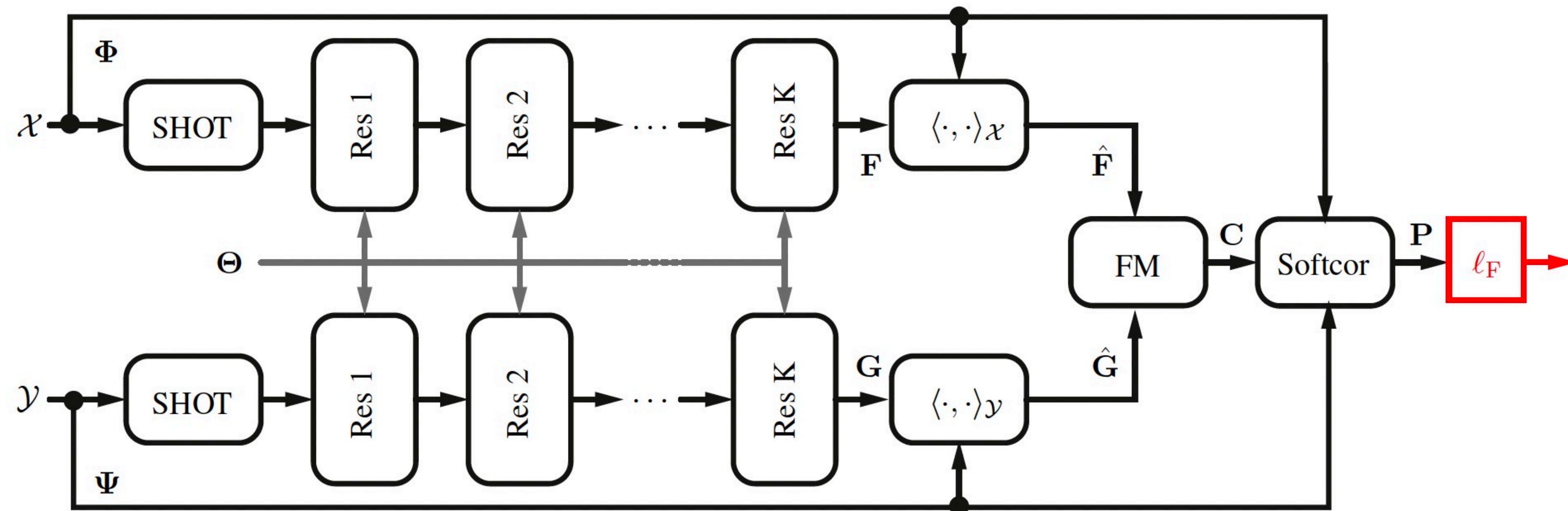


Input descriptors not in alignment

Results in poor texture transfer

FMNet

Main Idea : Learning approach

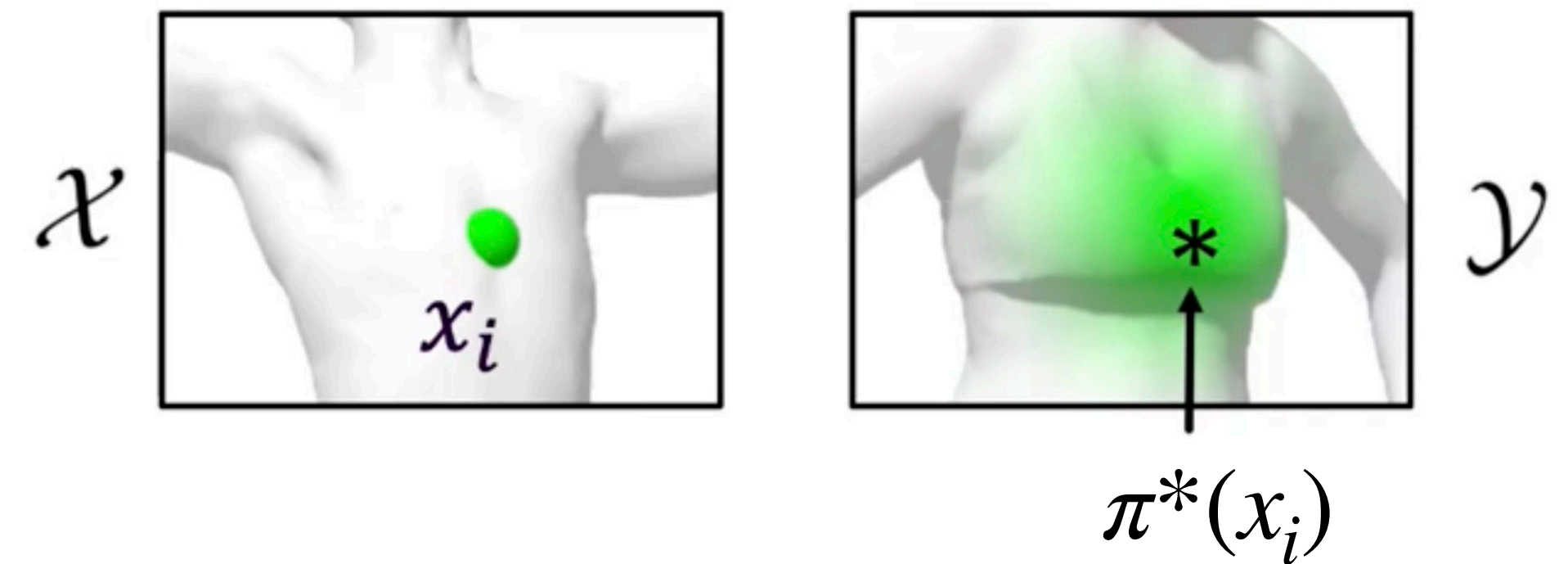
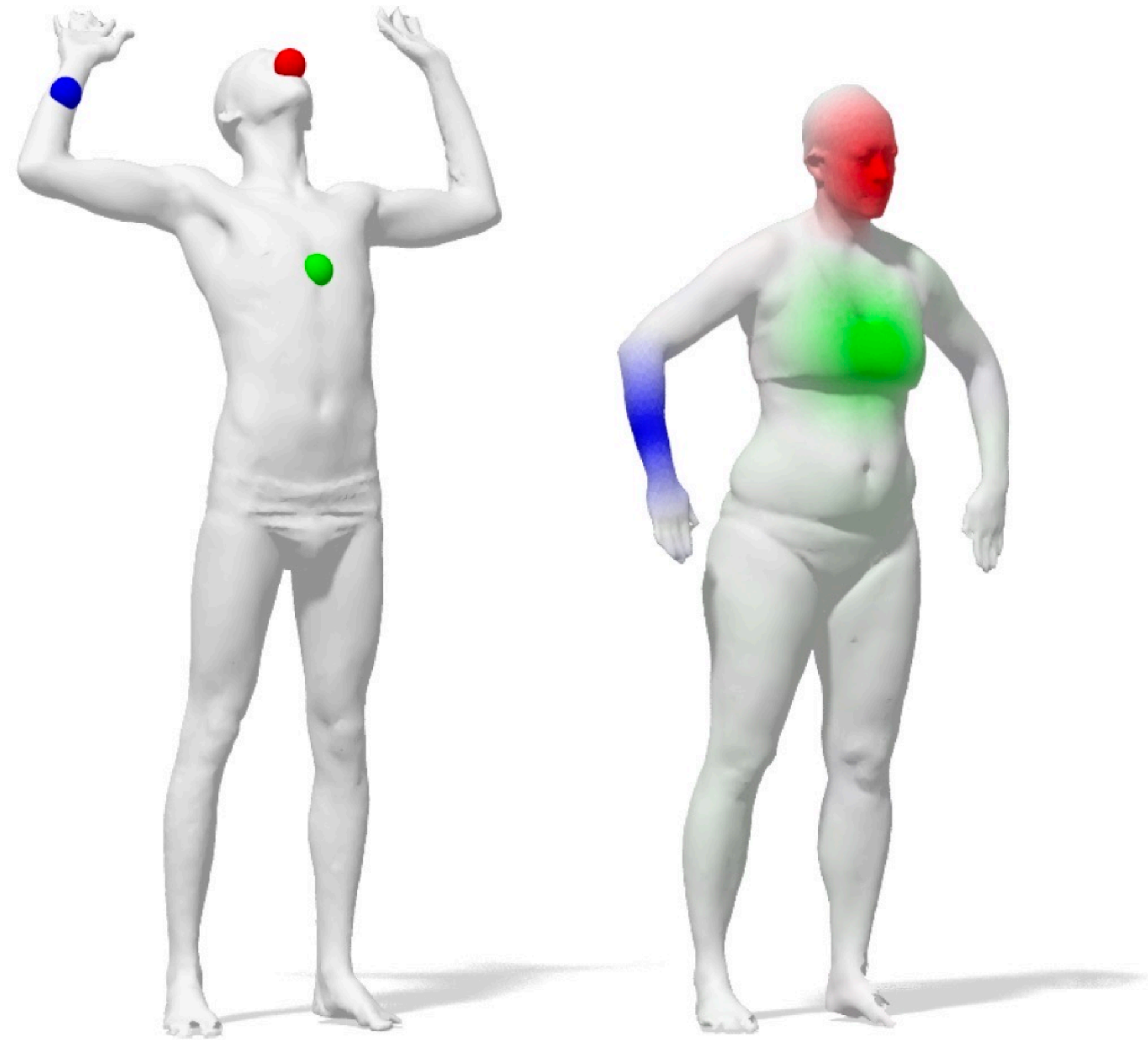


Training Loss :

$$\ell_F = \sum_{(x,y) \in (\mathcal{X}, \mathcal{Y})} P(x, y) d_{\mathcal{Y}}(y, \pi^*(x))$$

$P(x, y)$: soft map corresponding to C

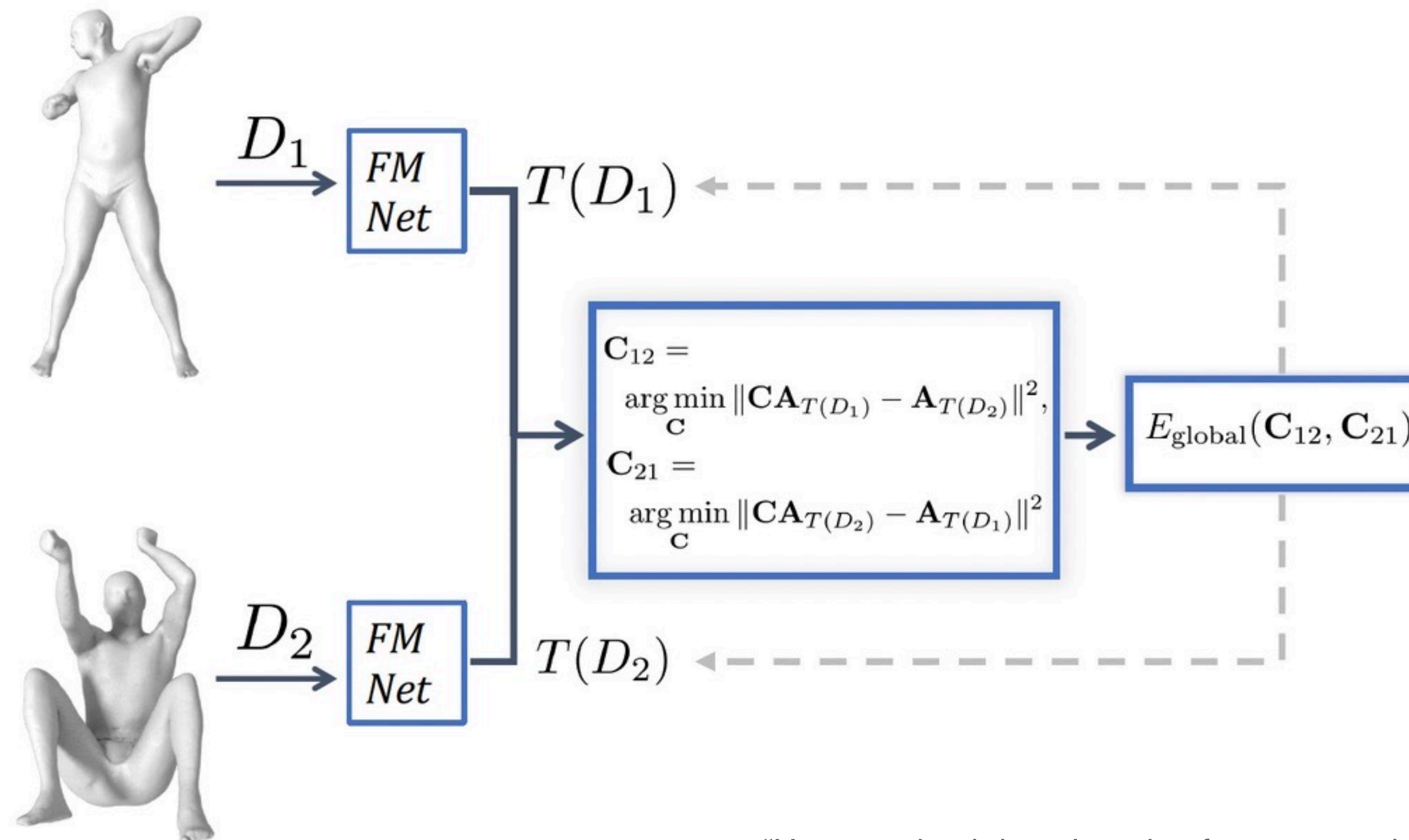
$$\ell_F = \sum_{(x,y) \in (\mathcal{X}, \mathcal{Y})} P(x, y) d_{\mathcal{Y}}(y, \pi^*(x))$$



Output as soft correspondence

SURFMNet

Main Idea : Make the loss fully unsupervised.



SURFMNet

$$Loss_{unsupervised} = \sum w_i E_i$$

$$E_1 = \|\mathbf{C}_{12}\mathbf{C}_{21} - \mathbf{I}\|^2 + \|\mathbf{C}_{21}\mathbf{C}_{12} - \mathbf{I}\|^2$$

Bijectionity

$$E_2 = \|\mathbf{C}_{12}^\top \mathbf{C}_{12} - \mathbf{I}\|^2 + \|\mathbf{C}_{21}^\top \mathbf{C}_{21} - \mathbf{I}\|^2$$

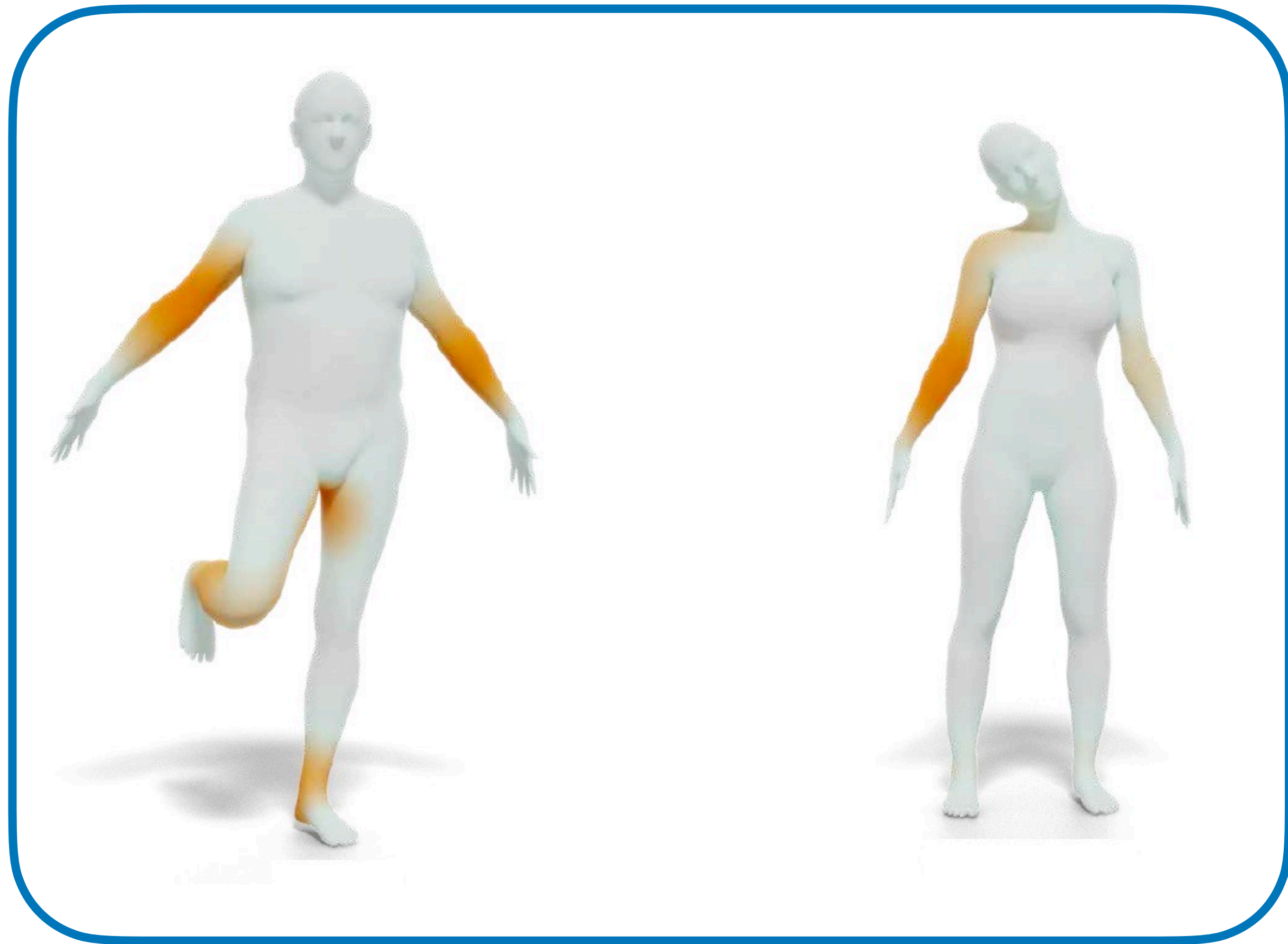
Area Preservation

$$E_3 = \|\mathbf{C}_{12}\mathbf{\Lambda}_1 - \mathbf{\Lambda}_2\mathbf{C}_{12}\|^2 + \|\mathbf{C}_{21}\mathbf{\Lambda}_2 - \mathbf{\Lambda}_1\mathbf{C}_{21}\|^2$$

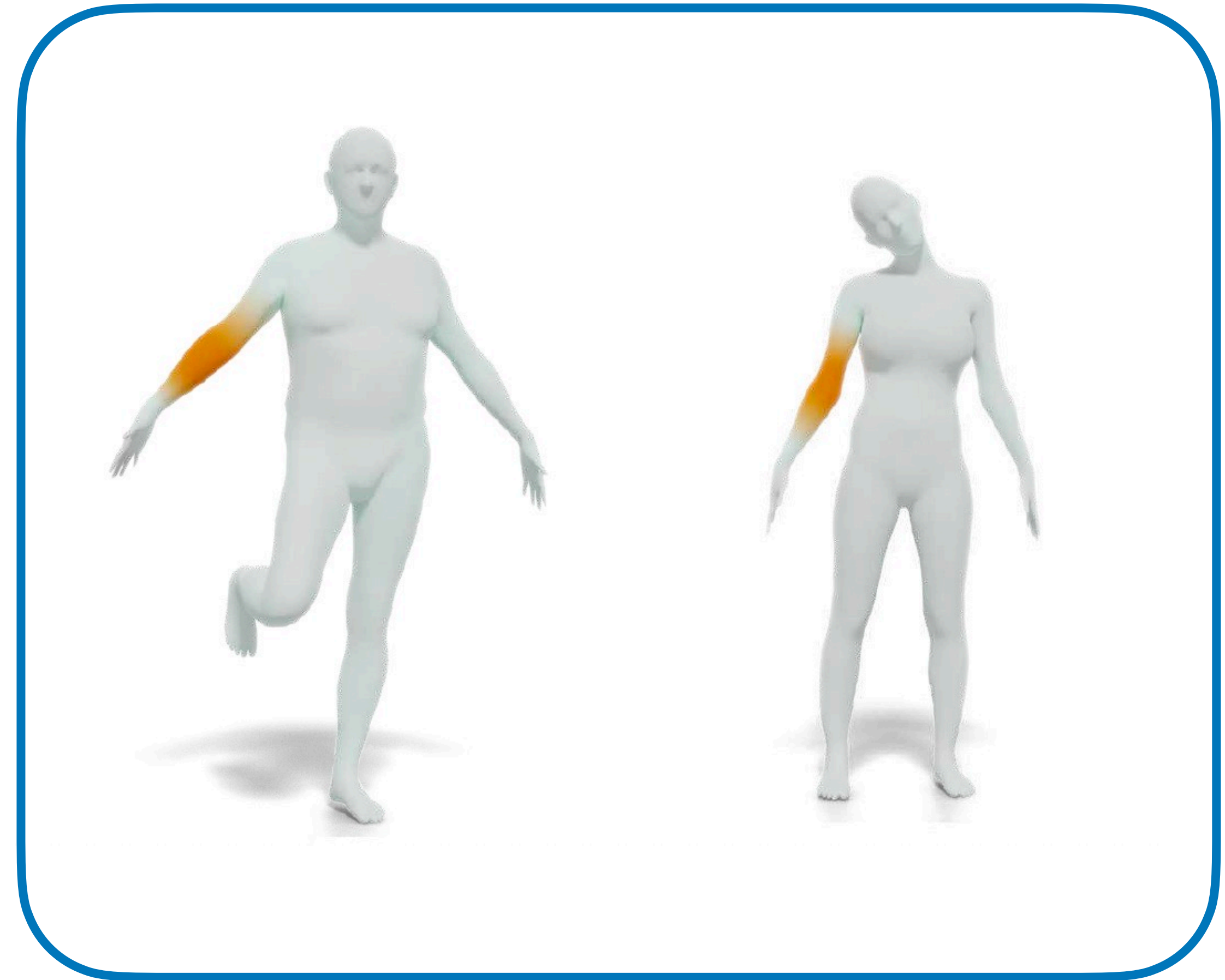
Near Isometry

All Penalties are in reduced basis. 50x faster than FMNet

Original vs Learned descriptors

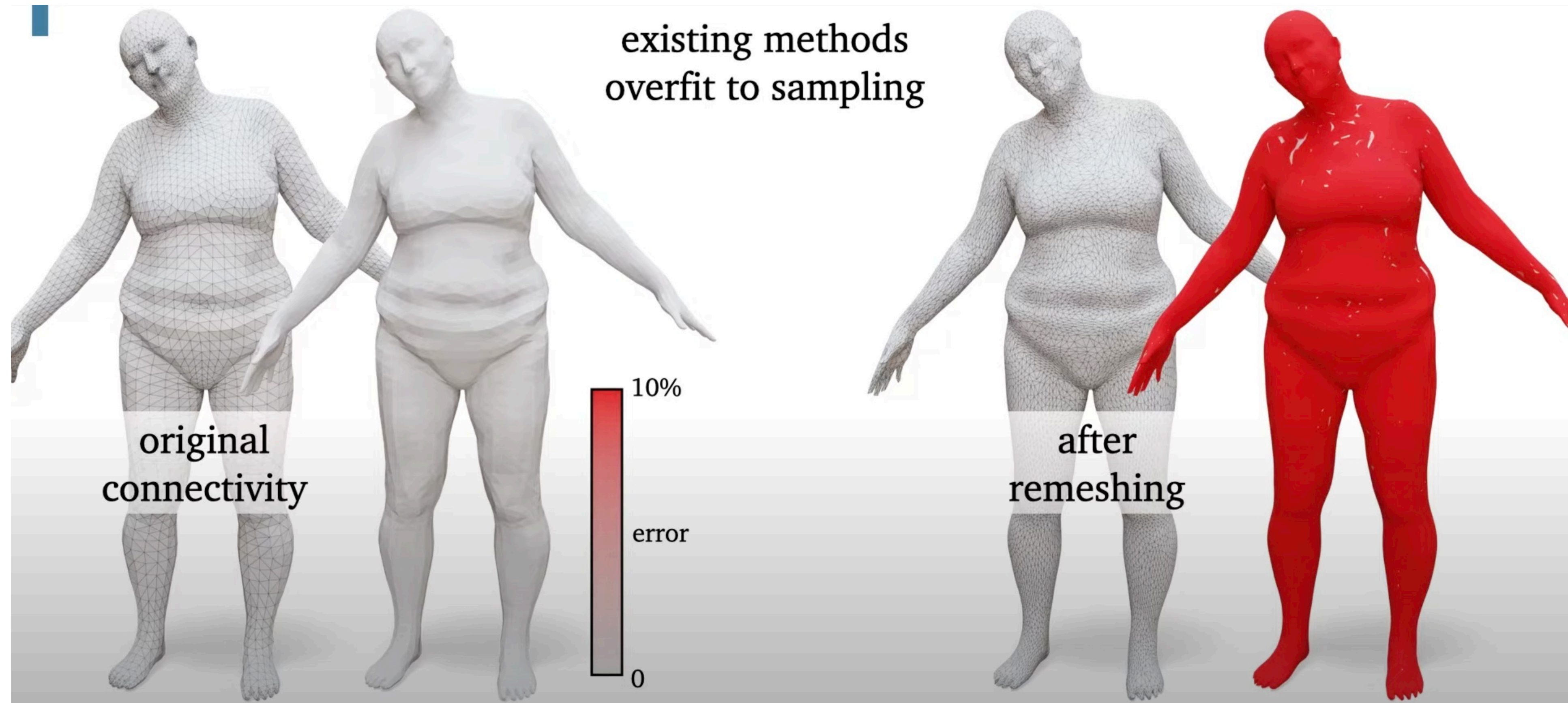


Before



After

Challenge : What happens under remeshing?



Diffusion based Network

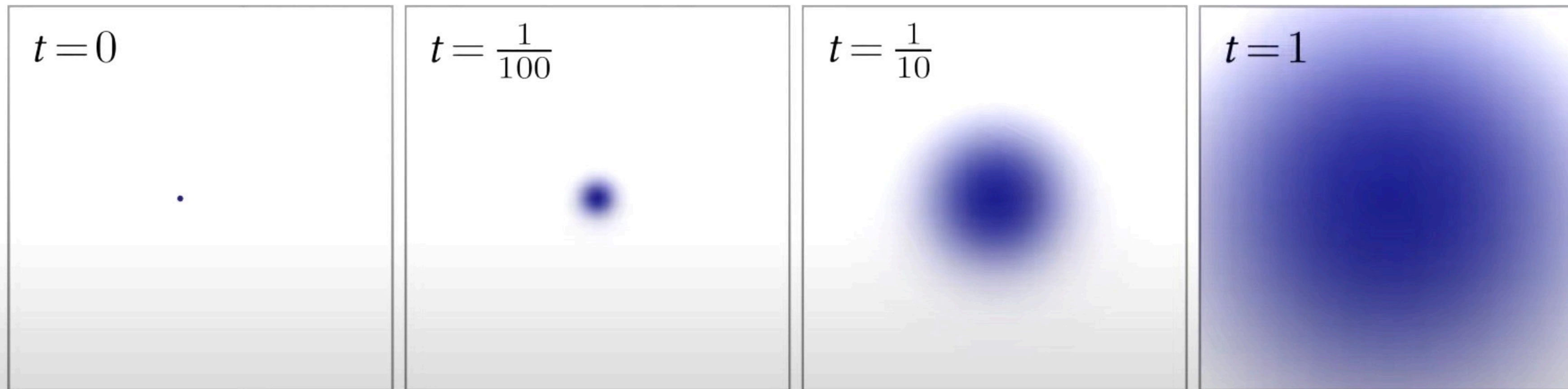


~~convolution
pooling~~ ✓
diffusion!

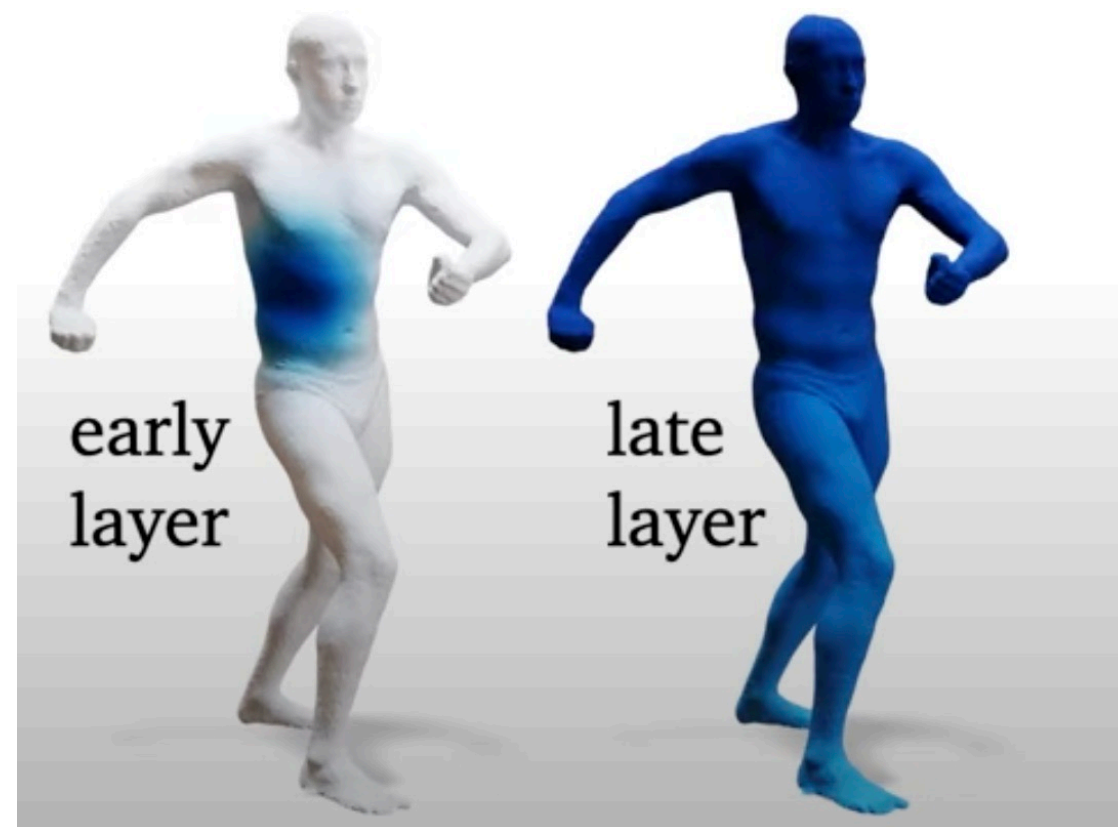
Recall : Laplacians and diffusion

$$\frac{d}{dt}x(t) = \Delta x(t)$$

—————diffusion of a point value—————→

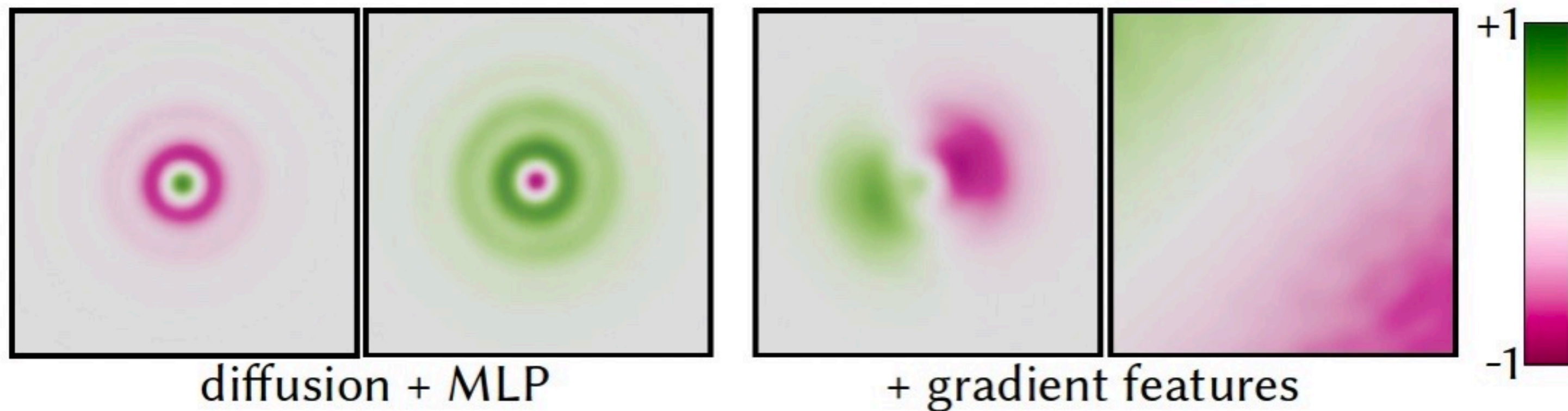


Learned diffusion & Spatial gradient

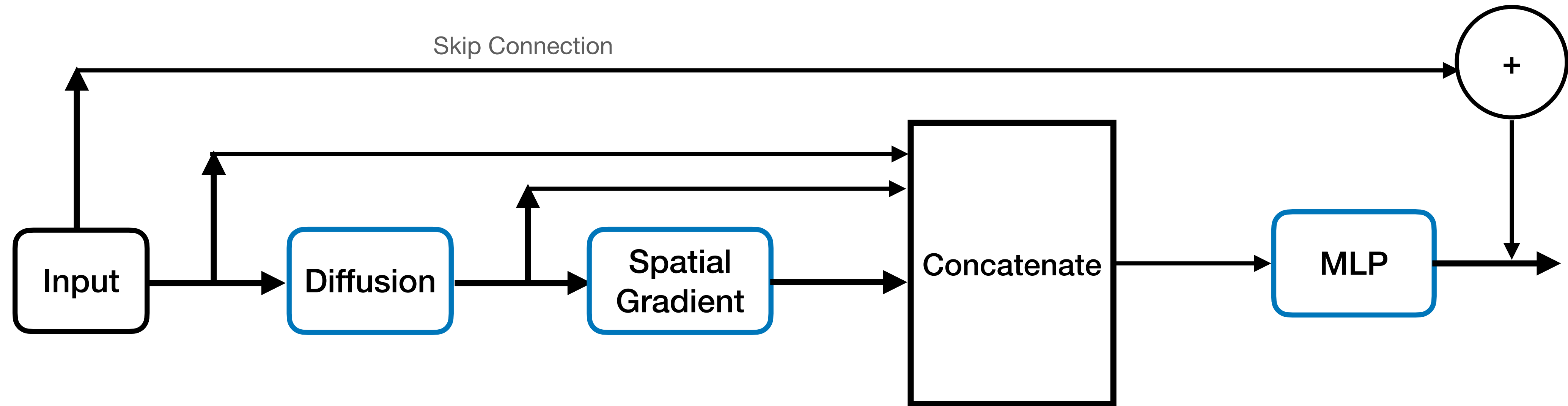


Key Idea : *diffusion time* is a learned parameter.

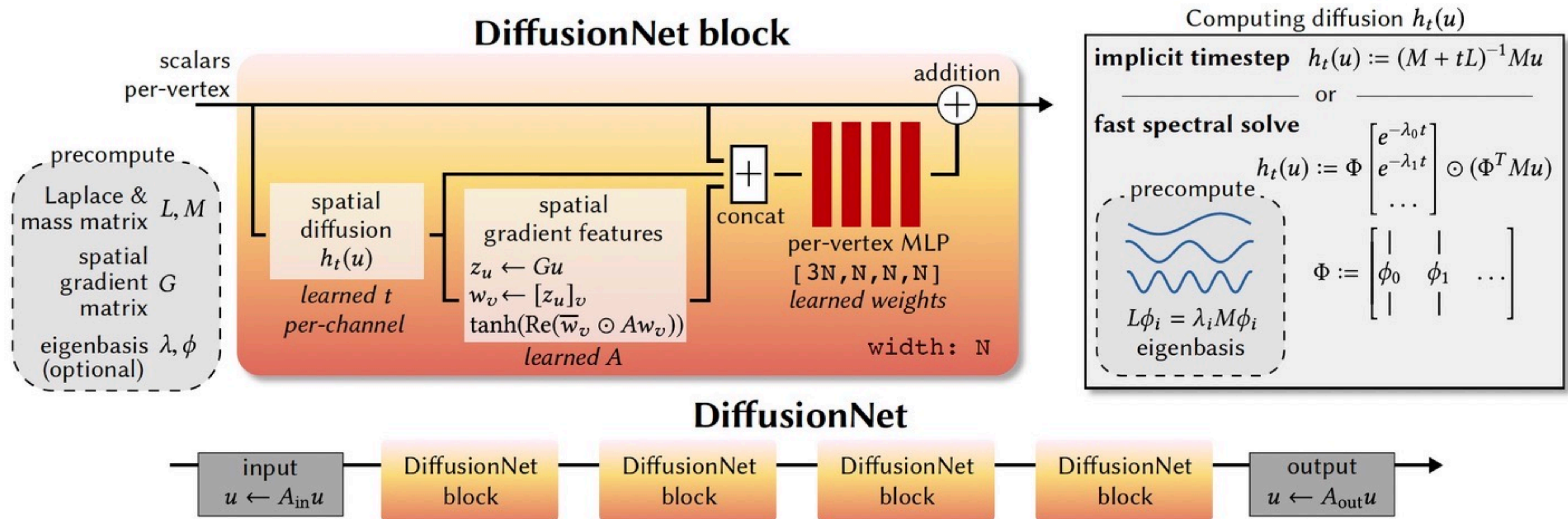
Lemma: diffusion + pointwise MLPs can represent all (radially symmetric) convolutions.



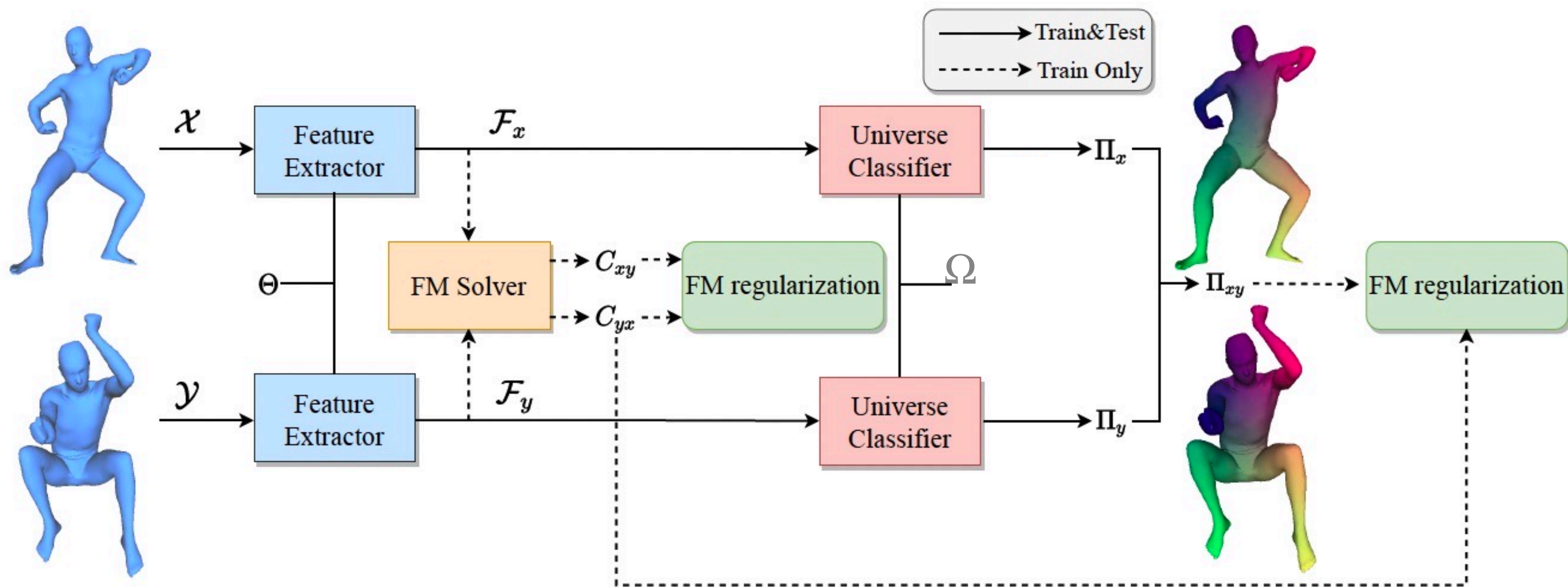
Diffusion block



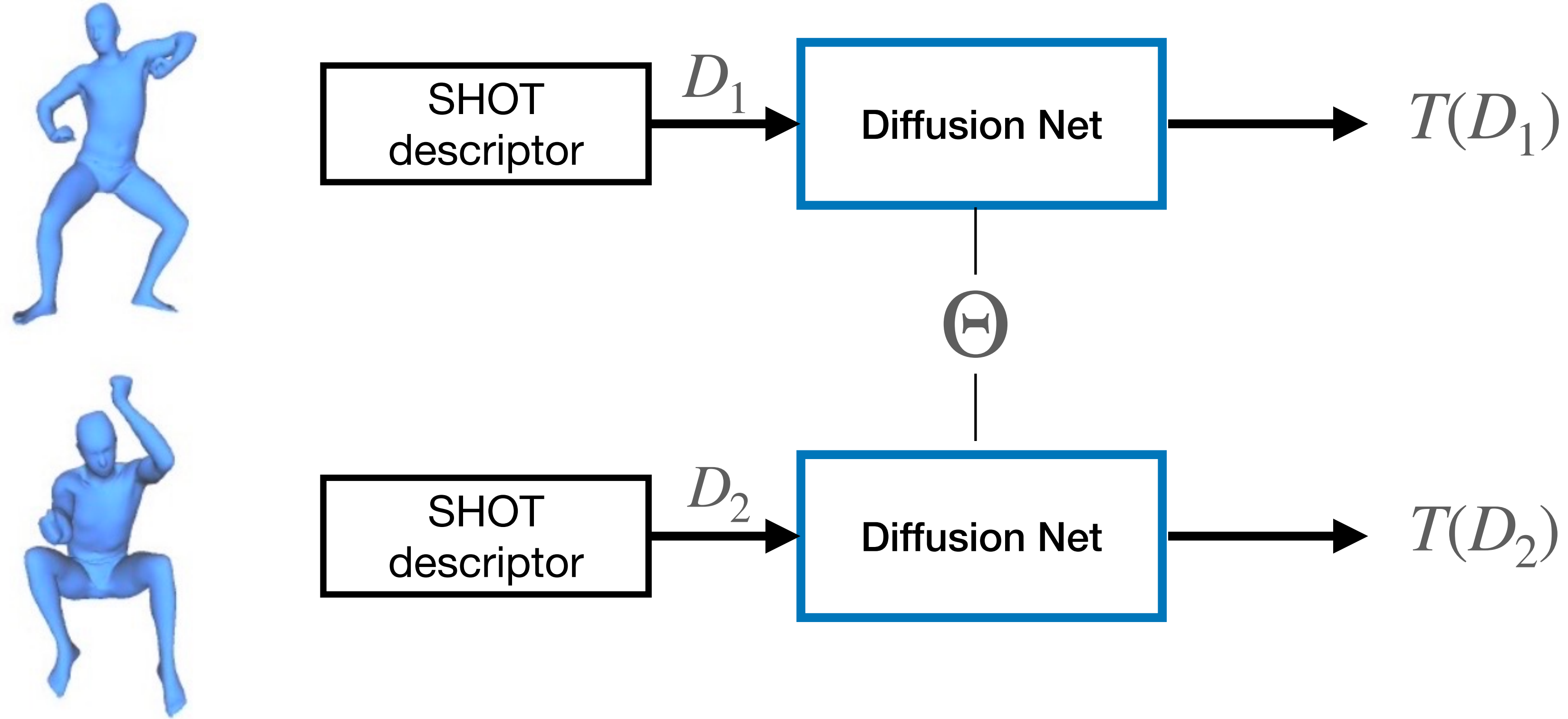
DiffusionNet



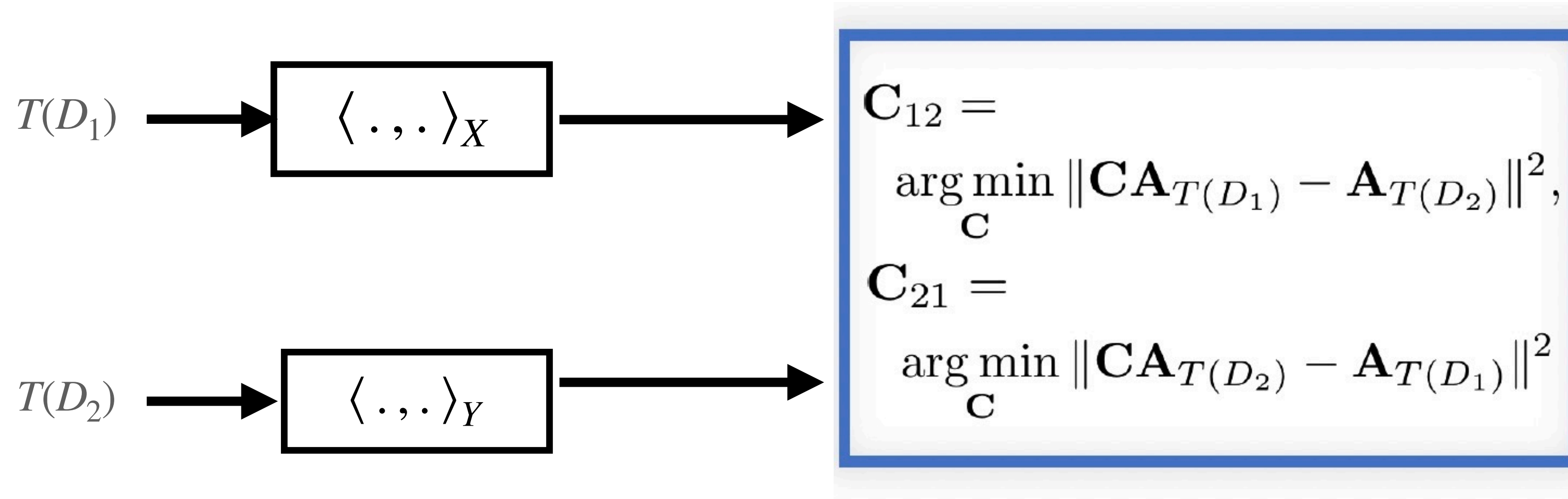
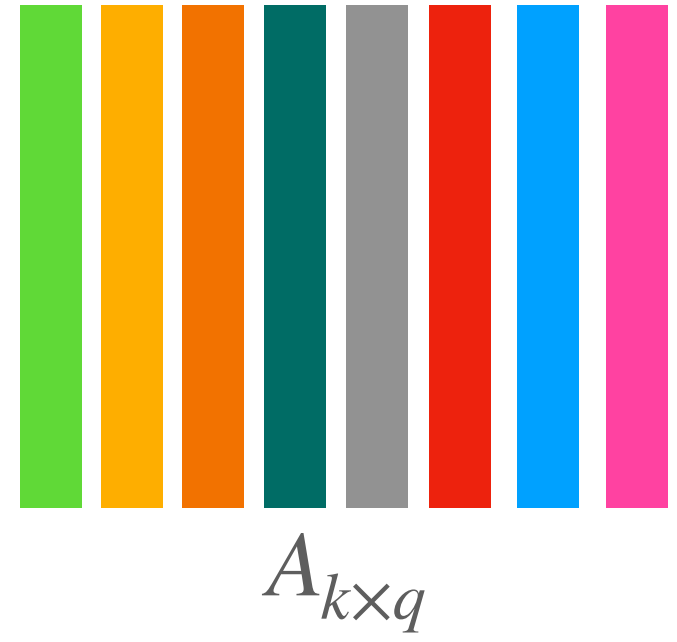
Our Unsupervised Deep Multi-shape Matching



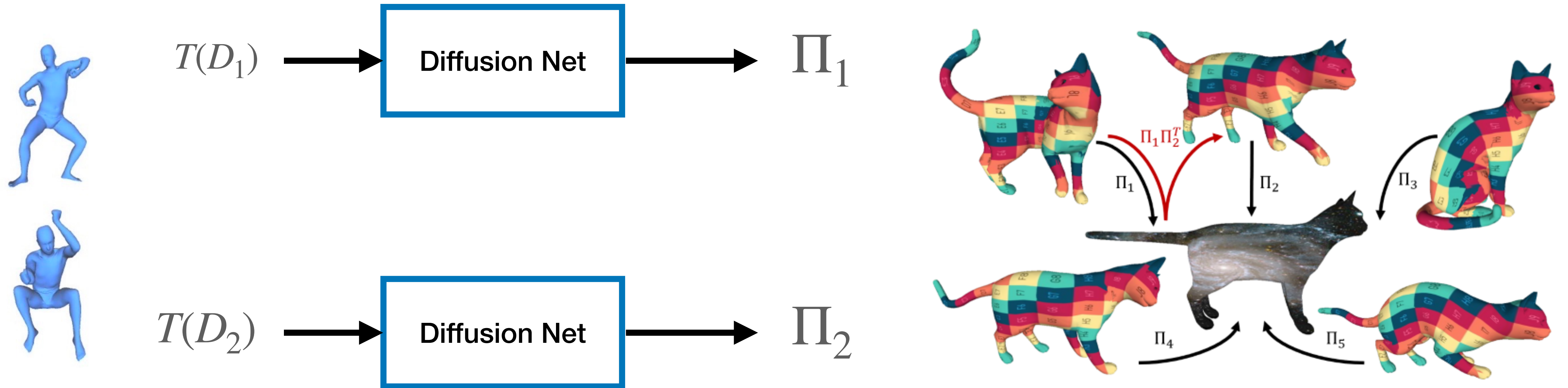
Feature Extractor



FM Solver



Universe Classifier



Unsupervised Loss

FM Regularisation

(Structural properties)



$$Loss_{total} = \sum w_i E_i$$

FM Regularisation

(Cyclic consistency)



$$E_1 = \|\mathbf{C}_{12}\mathbf{C}_{21} - \mathbf{I}\|^2 + \|\mathbf{C}_{21}\mathbf{C}_{12} - \mathbf{I}\|^2$$

Bijectivity

$$E_2 = \|\mathbf{C}_{12}^\top \mathbf{C}_{12} - \mathbf{I}\|^2 + \|\mathbf{C}_{21}^\top \mathbf{C}_{21} - \mathbf{I}\|^2$$

Area Preservation

$$E_3 = \|\mathbf{C}_{12}\mathbf{\Lambda}_1 - \mathbf{\Lambda}_2\mathbf{C}_{12}\|^2 + \|\mathbf{C}_{21}\mathbf{\Lambda}_2 - \mathbf{\Lambda}_1\mathbf{C}_{21}\|^2$$

Near Isometry

$$E_4 = \|\Phi_2 \mathbf{C}_{12} - \Pi_{21} \Phi_1\|^2 + \|\Phi_1 \mathbf{C}_{21} - \Pi_{12} \Phi_2\|^2$$

Experimental Results

Geodesic error(%)	FAUST	SCAPE	F on S	S on F
FMNet	11	17	30	33
SURFMNet	15	12	32	32
Ours	1.5	2.0	7.3	8.6

Ablation Study

Geodesic error(%)	FAUST	SCAPE	F on S	S on F
Classifier - free	2.1	3.8	17.4	22.9
Supervised	1.4	2.8	9.8	18.5
Ours	1.5	2.0	7.3	8.6

Conclusion

Functional Map Based

Unsupervised Learning of Feature descriptors

Cyclic Consistent